CERTAIN-COAST

Certified early warning to protect water, port and energy infrastructure on a sinking Adriatic coast

NATO SPS Multi-Year Project, call 2027-1 · Additional files

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The application form is self-contained; these files give detail for readers who want it. The technical note is background on the coupled model of work package 2 and on the placement of recharge and supply wells; its tests are synthetic and its numbers illustrative, not results at Durrës.

CERTAIN-COAST — Technical annex

Certified early warning to protect water, port and energy infrastructure on a sinking Adriatic coast

How the methods work, what was tested before the project, and what is new. This annex supplements the fields "How will the project reach the planned end state?" and "Work-plan" of the application form; work packages (WP), tasks (T), deliverables (D), months and milestones (MS) are those of the form. Test results come from synthetic test systems and preparatory models, not from measurements made by the project at the sites.

1. Two sites, three coupled hazards

The project protects two sites on the Albanian Adriatic coastal plain: at Durrës, the wellfields that supply drinking water and the port; near Fier, the landfall of the Trans Adriatic Pipeline (TAP). Both stand on low ground near active faults, and at Durrës the land is sinking; the earthquake of 26 November 2019 caused liquefaction near the coast (Mavroulis, Lekkas and Carydis, 2021). Three hazards threaten these assets, and they are coupled through two variables, the relative sea level and the water table. Sea level enters the aquifer at its seaward boundary and sets how far seawater advances; the water table sets the effective stress in the sands and so their resistance to liquefaction; subsidence raises the relative sea level at every asset. Pumping lowers the water table: it increases seawater intrusion and, below the preconsolidation head, permanent subsidence, but it raises the factor of safety against liquefaction. No hazard can be managed alone, so one indicator is defined per hazard, oriented so that larger means more dangerous (Table 1).

Table 1. Hazard indicators, thresholds and lead times.

Hazard (assets)	Indicator	Threshold	Lead time, cadence
Seawater intrusion (Durrës wellfields)	I_1: chloride at each production well	Agreed with the utility and AMBU (for example 250 mg/L)	Weeks to months; weekly
Liquefaction (Durrës port; TAP landfall)	I_2: liquefaction potential index under the design shaking at the current water table	Class boundary agreed with the operators	Daily; recomputed at once after an earthquake
Extreme water level (all assets)	I_3: total water level at each asset	Design or operational level of each asset	Hours to days for storms; return levels yearly

An alarm is missed when the indicator exceeds its threshold within the lead time and no alarm was issued; the long-run miss rate is the quantity the project certifies. Earthquakes are not forecast: the liquefaction alarm states how the ground would respond if the design earthquake struck now, a state that changes with the water table.

2. The method chain

Six steps turn data into bulletins (Figure 1). Each step has a pass test and a fallback (Section 7). Three loops close the chain: verified outcomes update the alarm margins, new data are assimilated into the model, and the certified safe window of Section 6 tells the network how many years of safe operation each sensor adds.

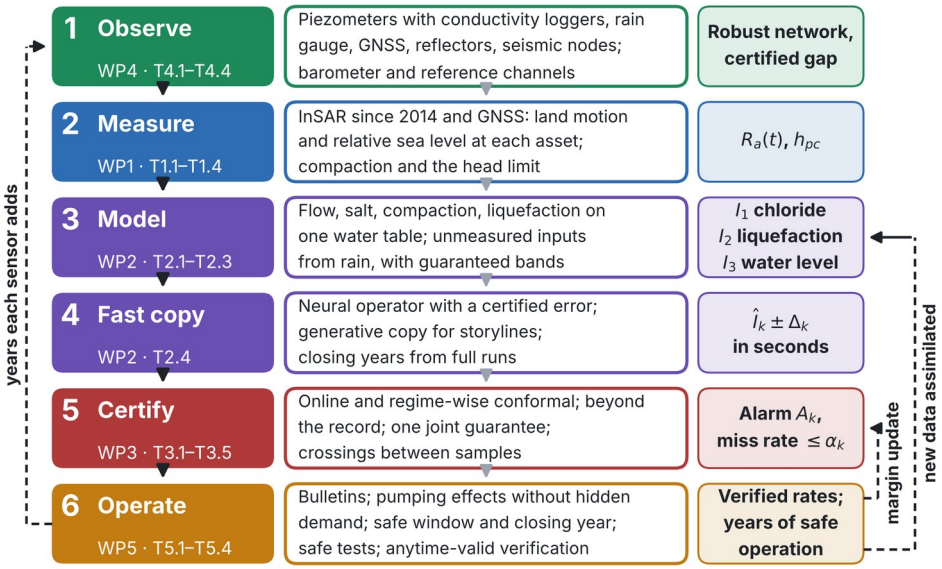


Figure 1. The CERTAIN-COAST method chain for the Durrës wellfields and port and the TAP landfall near Fier. Solid arrows: from data to bulletins. Dashed arrows: the three feedback loops.

3. Observe and measure (WP1, WP4)

3.1 Relative sea level at each asset (WP1, T1.1–T1.4)

Sentinel-1 ascending and descending time series since 2014 give the vertical and east components of land motion; a trajectory model separates trend, season, the coseismic step of 26 November 2019 and the postseismic transient. Corner reflectors and GNSS at the assets (T1.2) give stable radar targets on soft ground and tie the solution to a global frame. The tide gauge records sea level η_{TG} relative to its own benchmark, which also moves, so the relative sea level at asset a adds the differential vertical land motion u_z, propagated with its full covariance (target vertical-velocity uncertainty 1–2 mm/yr; Ohenhen et al., 2023):

$$R_a(t) = \eta_{\mathrm{TG}}(t) + u_z^{\mathrm{TG}}(t) - u_z^{a}(t) \tag{1}$$

Elastic and inelastic compaction fitted to the InSAR and head records give the head limit h_{pc} above which pumping causes no new permanent subsidence (Lees, Knight and Smith, 2022). The same $R_a(t)$ drives all three hazard models, so the warnings cannot disagree about where the sea is.

3.2 A network designed for the warnings, which corrects its own sensors (WP4, T4.1–T4.4)

The network is designed before it is bought (D4.1, Month 6). Candidates are piezometers with conductivity loggers, a local rain gauge, GNSS, corner reflectors, seismic nodes and, where access allows, fibre-optic sensing; the monitoring wells of AMBU (existing wells only) are candidate sites. Each candidate is valued by goal-oriented Bayesian design (Attia, Alexanderian and Saibaba, 2018) in two currencies: the guaranteed width it removes from the warning bands, and the years of certified safe operation it adds (Eq. (11)). A sensor is bought only if its guaranteed value is positive: in a preparatory test, a tide gauge that a standard ensemble rated as narrowing the recharge band by 15% widened the guaranteed band by 16–21%.

Loggers respond to air pressure (Rasmussen and Crawford, 1997) and drift. A barometer and a sealed reference channel on each logger hub are therefore design candidates beside the piezometers; their errors are removed by marginalisation, and the network is chosen against the worst plausible error model, with a certified distance from the best network (robust design; Chowdhary, Attia and Alexanderian, 2025). In a preparatory test with 12 sensors the worst-case information was 5.59 nats against 3.67 for the classical design (+52%), with a certified optimality gap of 18%; 90% forecast intervals covered 88–93% of outcomes after correction and 49–62% without, and corrected heads were within 2.9 cm root-mean-square against 7.6 cm raw.

At the TAP landfall, T4.2 installs one GNSS station, two corner reflectors, a seismic station and a water-table logger under TAP's security rules. Ambient seismic noise tracks the water table (Clements and Denolle, 2018), also on unused telecommunication fibre (Rodríguez Tribaldos and Ajo-Franklin, 2021); once calibrated against piezometers (T4.4), velocity changes measure the water table between wells at the port and the landfall, and a drop after an earthquake measures the softening of the ground.

4. Model and certify (WP2)

4.1 One coupled model (T2.1–T2.3)

One model, set up for the two sites on one coastal motion field (T2.1), couples variable-density flow, salt transport with memory in clays, the sea as a moving boundary driven by $R_a(t)$, compaction, the total water level at each asset, and liquefaction triggering with the water table taken from the flow model at every time step (Seed and Idriss, 1971; Boulanger and Idriss, 2014). Groundwater rise under sea-level rise is a hazard in its own right (Befus et al., 2020) and raises liquefaction severity where water tables are shallow (Grant et al., 2021); here one water table drives all three hazards, so a pumping decision is checked against all of them at once. Parameters are estimated jointly from heads, conductivity, chloride, InSAR and seismic velocity by an ensemble smoother (Emerick and Reynolds, 2013); the model is differentiable, so adjoint gradients of every indicator are available (Shen et al., 2023). Priors are stated so that an expert can read and dispute them (Simpson et al., 2017; Fuglstad et al., 2019) and are checked after calibration for prior-data conflict (Evans and Moshonov, 2006).

4.2 Guarantees for the inputs and boundaries nobody measures (T2.3)

Recharge, vadose delay, storage, the sea-level and inland boundaries, the initial head and the dispersion width are never measured, yet every warning depends on them. They are reconstructed from rain and heads by a function-space diffusion prior with an ensemble update through the exact solver, with the errors of the rain product simulated in the prior. Their bands are calibrated by conformal risk control (Angelopoulos, Bates, Fisch, Lei and Schuster, 2024) on simulated aquifers that carry these errors, and checked on measured anchor periods, so the guarantee does not depend on the simulator. With a loss $L_i(\lambda)$ between 0 and 1 for calibration case i and its mean $\hat{R}_n(\lambda)$ over n cases:

$$\hat{\lambda} = \inf\left\{\lambda : \frac{n}{n+1}\hat{R}_n(\lambda) + \frac{1}{n+1} \le \alpha\right\} \text{ and } \mathrm{E}\left[L_{n+1}(\hat{\lambda})\right] \le \alpha \tag{2}$$

For α = 10%, at least 9 anchor periods give the average guarantee and 22 a 90% certificate. Under drift the calibration is matched to the forecast horizon, and a computable floor shows how far each band is from the narrowest valid one. In the preparatory test the raw ensemble missed 17–20% of the weeks and the calibrated bands 9.5–10% (target 10%); the risk stayed at 0.003–0.082 up to 20 years ahead under drift.

4.3 The boundary where the model is cut (T2.3, T4.1)

Every model of the aquifer ends somewhere inland, at a cut that geology does not mark. A fixed head, a prescribed inflow or a general-head boundary does not respond to pumping or to the sea as the aquifer beyond does, and the choice decides the answers. In the preparatory test, the fresh water lost to a 0.5 m rise of the sea over 20 years was 49 m³ per metre of coast with a prescribed inflow, 462 with a fixed head and 154 in the full aquifer (Werner and Simmons, 2009). For any linear aquifer beyond the cut, heterogeneous, leaky or closed, the exact boundary is a conductance plus positive storage compartments. With $u = h - \acute{h}$ the head at the cut above its reference and q_{in} the inflow from rain, the outward flux is

$$q_n = -q_{in}(t) + C\,u + \sum_{m=1}^{M} w_m (u - y_m) \text{ and } \dot{y}_m = \mu_m (u - y_m) \text{ and } C, w_m \ge 0 \tag{3}$$

Conversely, every such network is the exact boundary of some physical aquifer, so a learned boundary cannot be unphysical; the same positive memory describes diffusion into the silt and clay lenses inside the model (Haggerty and Gorelick, 1995). With $C, w_m \ge 0$ the boundary is passive: the model loses energy whatever its other parameters, as passive models of an external electrical network do (Gustavsen and Semlyen, 2001). A negative conductance that was stable in a 2 km test grew without bound in a 6 km model. The rates μ_m sit on a fixed logarithmic grid, three to four per decade of period. The weights are learned from two piezometers between the wellfields and the cut (T4.1), which separate the boundary from everything else in the model, on the band of periods that changes of pumping excite. A neural network may supply q_{in} from rain and monotone conductances, never the storages, and a prior learned from simulated geology fills the periods the data do not reach. In the test, six compartments reproduced the full aquifer to 7.8 mm root-mean-square over 20 years (best general-head boundary: 148 mm). Learned from the two piezometers while the pumping varied, the boundary predicted an unseen drought with doubled pumping to 6 mm (372 mm), and the fresh water lost to the sea-level rise within 4%.

4.4 A fast copy with a certified error (T2.4)

Warnings need thousands of model runs a day. A neural operator (Li et al., 2021; Lu et al., 2021) is trained on runs of the coupled model, and its error on each forecast is bounded conformally (Ma et al., 2024), with the score normalised by the goal-oriented dual-weighted residual η_k (Becker and Rannacher, 2001), so the bound is tight where the copy is right about the quantity that triggers the alarm:

$$\Delta_k(\xi) = \hat{q}_{1-\delta}\,\eta_k(\xi) \text{ and } Pr\left[\left|I_k(\xi) - \hat{I}_k(\xi)\right| \le \Delta_k(\xi)\right] \ge 1 - \delta \tag{4}$$

Forecasts whose bound is too wide are recomputed with the full model on a reduced domain. A generative copy (Gao, Kaltenbach and Koumoutsakos, 2024) explores long storylines of sea-level rise, subsidence, droughts and pumping and chooses which of them the full model runs; in a preparatory test its statistics alone put the closing year of the safe window 2–8 years too late, so closing years are always certified from full runs with the exact binomial bound (Clopper and Pearson, 1934).

5. Certified alarms (WP3)

5.1 Alarm rule, calibration under shift and beyond the record (T3.1–T3.2)

Thresholds τ_k and tolerable miss rates α_k are agreed with the operators (T3.1). Alarm k fires when the forecast at lead time L_k comes within a margin m_k of its threshold. The margin adds the certified copy error to a forecast error γ_k that rises after a miss and falls slightly otherwise; with a constant step λ the long-run miss rate stays within $C_k/(\lambda T)$ of its target for any sequence of data (Gibbs and Candès, 2021, 2024; Angelopoulos, Barber and Bates, 2024):

$$A_k(t) = 1\left\{\hat{I}_k(t+L_k) \geq \tau_k - m_k(t)\right\} \text{ and } m_k(t) = \Delta_k(t) + \gamma_k(t) \tag{5}$$

$$\gamma_k(t+1) = \gamma_k(t) + \lambda_t\left[\text{miss}_k(t) - \alpha_k\right] \text{and} \left|\frac{1}{T}\sum_{t=1}^{T} \text{miss}_k(t) - \alpha_k\right| \leq \frac{C_k}{\lambda T} \tag{6}$$

The guarantee also holds within each regime (drought, storm season, after an earthquake, each asset) by conformal prediction with conditional guarantees (Gibbs, Cherian and Candès, 2025), and for events rarer than the record by extreme conformal prediction (Pasche, Lam and Engelke, 2026). Still-water extremes use a generalised Pareto tail (Coles, 2001) whose threshold moves with the measured relative sea level of Eq. (1). In preparatory tests on a coupled variable-density model, an extreme-aware margin kept the share of seasons above the chloride limit at 4–9% across drought, human-feedback and compound storylines, where a static alarm allowed 13–21%.

5.2 One guarantee for the three assets (T3.3)

The alarms share inputs, so their misses are dependent. The probability that any alarm is missed is bounded by the sum of the tolerances, which are allocated where tightening them costs least in consequence-weighted false alarms, within caps agreed with the operators. The margins are calibrated together by Learn then Test (Angelopoulos, Bates, Candès, Jordan and Lei, 2025), so every certified rate holds at once with probability at least $1-\beta$. Civil protection receives one joint guarantee for the wellfields, the port and the landfall.

5.3 Warnings for what happens between samples (T3.2)

Chloride is sampled weeks or months apart. Sparse samples pin the distribution of chloride on the sampling dates but not its time scale, so a well can turn salty between two samples that both look safe. The warning is therefore the probability that chloride crosses its limit $c^\star$ at any time in the season T, reported as a band over the set M of all models consistent with the records, in the spirit of robust bounds over calibrated models (Gierjatowicz et al., 2020) and of online forecasts from irregular records (Krach, Nübel and Teichmann, 2026):

$$\underline{p} = \inf_{P \in M} P\left[\max_{t \in T} c(t) > c^\star\right] \text{ and } \overline{p} = \sup_{P \in M} P\left[\max_{t \in T} c(t) > c^\star\right] \tag{7}$$

The band is widened by a conformal margin calibrated on simulated aquifers and anchor periods and issued online as the conditional probability given the irregular samples and the daily heads; the network design adds the observation that closes it (a logger, or samples paired 1, 3 and 7 days apart). In the preparatory test the chance of a crossing within the season was 0.78, against 0.023 on a sampling date (34 times); the band was 0.18–1 with quarterly samples and 0.76–0.80 with a daily logger; for skewed chloride the margin restored 90.3% coverage (74% without), and a head logger raised the warning skill from 7% to 20%.

5.4 Back-testing and verification in operation (T3.4–T3.5)

The chain is replayed over the record since 2014 with the system frozen at each issue time, under a protocol published with a DOI before the results (D3.2, Month 27). At 95% confidence, verifying a 10% miss rate needs 29 exceedance events with no miss, 46 with one or 61 with two; a 5% rate needs 59, 93 or 124. In operation the ledger is published every quarter, so each miss rate is looked at many times, and a fixed-sample bound recomputed at every look loses its guarantee: in a simulation of six calibrated alarms it flagged at least one as broken in 68% of runs. Verification therefore uses e-processes, which stay valid however often they are checked (Ville, 1939; Vovk and Wang, 2021; Ramdas et al., 2023), with $x_i = 1$ if event i was missed:

$$E_k(n) = \sum_j \pi_j \prod_{i=1}^{n} \left(\frac{q_j}{\alpha_k}\right)^{x_i} \left(\frac{1-q_j}{1-\alpha_k}\right)^{1-x_i} \text{ and } Pr\left(E_k(n) \geq 1/\beta \text{ for some } n\right) \leq \beta \tag{8}$$

In the simulation the e-process flagged a calibrated alarm in 2.2% of runs (at most 5% is guaranteed) and detected an alarm whose miss rate had risen to 25% in 84% of runs within 100 events.

6. Operate and decide (WP5)

Twelve months of live bulletins (T5.2) follow a dry run (T5.1). Each bulletin states, per asset, the forecast, the margin, the alarm state, the certified miss rate and the distance to the closing of the safe window, and is logged with its inputs and code version so that it can be reproduced and audited. A post-event protocol agreed with civil protection (T5.4) remeasures heads, chloride, GNSS and seismic velocity within days of a strong earthquake or storm, updates the model and recomputes the alarms and the window.

6.1 The effect of pumping without the hidden drought demand (T5.3)

In droughts the utility pumps more while recharge falls, so a hidden cause drives both the pumping and the salinity, and substitutes for hidden confounders inferred from the treatment history (Bica, Alaa and van der Schaar, 2020) can miss it. CERTAIN-COAST measures the confounding path instead: the heads, after the drawdown measured in the pumping tests is added back, read the recharge, and a calibrated physics check rejects any substitute that misses a confounder before an effect is reported. In the preparatory test the bias of the summed well effects was 0.59 unadjusted, 0.56 with a history substitute and 0.00 with the heads; the check had a size of 6% and a power of 99–100%. On the calibrated chloride model of Fushë-Kuqe, 40% less recharge would grow the shallow area above 250 mg/L from 31 to 65 km² (effect +33 km², conformal band 15–49 km²).

6.2 The certified safe operating window and the year it closes (T5.3)

Pumping moves the hazards in opposite directions, and sea-level rise and subsidence tighten all of them. At each bulletin and over the adaptation horizon the project computes the set of decisions u (wellfield pumping Q_w, port dewatering) that meets the demand D, keeps every hazard within its certified limit and causes no new permanent subsidence, and the relative sea level and year at which this set becomes empty:

$$W_\alpha(R) = \left\{ u : Q_w \geq D,\ \Pr_\theta\left[I_k(u, R; \theta) > \tau_k\right] \leq \alpha_k \left(k = 1,2,3\right),\ h(x, u, R) \geq h_{\mathrm{pc}}(x) \right\} \tag{9}$$

$$R^\star_\alpha = \text{largest } R \text{ with } W_\alpha(R) \neq \varnothing \text{ and } t^\star_\alpha = \text{first } t \text{ with } W_\alpha\left(R(t)\right) = \varnothing \tag{10}$$

This is a certified, multi-hazard adaptation tipping point (Kwadijk et al., 2010; Haasnoot et al., 2013); tipping points exist for seawater intrusion alone (Mazi, Koussis and Destouni, 2013). At the closing point the least dewatering that keeps liquefaction within its limit, A, meets the most dewatering the wellfield tolerates, B, and one multiplier μ converts any change of a margin, from a sensor, a better model or a pumping test, into years:

$$\mu = -\left[\frac{d(B-A)}{dt}\right]^{-1}_{t=t^\star_\alpha} \text{ and } \Delta t^\star_\alpha \approx \mu\left(\Delta B - \Delta A\right) \tag{11}$$

On a synthetic transect with the three hazards of Durrës and illustrative values, the certified window closed at a relative sea level of 0.41 m, after 38 years; plans that watch one hazard believed they had 43 to 97 years, and the margins cost 84 years against the window without them. Valued in years by preposterior analysis, with Eq. (11) as a first-order shortcut, a sentinel piezometer between the wellfield and the sea added 21.6 years, and a GNSS station and two corner reflectors 5.6 and 5.0 years, where the classical criterion values them at zero because they narrow none of today's forecasts.

6.3 Safe pumping tests chosen for the years they add (T5.3)

Routine pumping keeps the water table in a narrow band, so the records say little about how chloride responds near the onset of intrusion, which decides when the window closes. Short pumping and dewatering tests, each approved by the utility, approach that onset safely. The response is linear in unknown coefficients of hinge features of the head, with a confidence set that holds in every week (Abbasi-Yadkori, Pál and Szepesvári, 2011). A test is chosen for the years it adds to the certified closing year, as in active identification with guarantees (Mania, Jordan and Recht, 2022) but for the decision, and only if the upper confidence bound of chloride stays below a test limit in every week of the test and the recovery (Koller et al., 2018). In the preparatory test the campaign raised the certified closing year from 10.8 to 31.9 years (true value 45) with chloride below 123 mg/L in 95% of runs, whereas tests chosen for parameter accuracy alone passed 250 mg/L in 42% of runs; the certificate held in all 1,000 runs. The years rest on the prior from the calibrated model, so the certified year is reported for the prior error radius S and for $2S$ (doubling it cost 17.3 of the 31.9 years), for readers who trust the model less (Andrews and Shapiro, 2021).

6.4 Pumping rule and the value of the warning for each user (T5.3, D5.1)

Within the window, the pumping and drought rule supplies as much water as possible; it is checked for a sudden pumping stop, after which the water table recovers and the factor of safety falls, computed with the fast copy, confirmed with the full model and agreed with the utility and AMBU. Because each miss rate is certified, the value of a warning to each user in the cost-loss model (Murphy, 1977; Richardson, 2000) has a guaranteed lower bound for every cost-loss ratio agreed in T3.1, reported in D5.1.

6.5 Pumping before the source of the chloride is known (T4.1, T5.3)

Rising chloride at a wellfield need not be seawater. Deep brine drawn up under the wells, a drought that concentrates the recharge, or irrigation return flow and wastewater fit the same chloride record, and each calls for a different change of pumping. On a synthetic test bed with three wellfields and an alternative supply, acting on the wrong source let the blend exceed 250 mg/L with probability up to 100%, and the usual assumption of seawater intrusion up to 98%. CERTAIN-COAST therefore decides before the source is known. Tracer data y (Br/Cl ratios, water isotopes, depth profiles from multilevel wells, nitrate and boron; Davis, Whittemore and Fabryka-Martin, 1998) give each candidate source k an e-value. The sources that the data cannot exclude form a set that contains the true source with probability at least $1-\beta$, for any sample size and without a prior (Vovk and Wang, 2021):

$$E_k(y) = \frac{1}{K-1}\sum_{j \neq k} \frac{p_j(y)}{p_k(y)} \quad \text{and} \quad S(y) = \{k : E_k(y) < 1/\beta\} \quad \text{and} \quad \Pr_k\left[k \notin S(y)\right] \le \beta \tag{12}$$

The set keeps every source whose posterior under a flat prior exceeds $\beta/(K-1+\beta)$, and it is never empty. Where the end-member signatures are uncertain, the denominator is replaced by its supremum over them (universal inference; Wasserman, Ramdas and Balakrishnan, 2020). The pumping of the wellfields and the use of the alternative supply are then chosen at least social cost, safe for every source left in the set:

$$\min_u c(u) \ \text{s.t.} \ \Pr_k\left[\mathrm{Cl}_{\mathrm{blend}}(u) > 250\,\mathrm{mg/L}\right] \le \alpha \ \text{for every} \ k \in S(y) \Rightarrow Pr\left[\text{exceed}\right] \le \alpha + \beta \tag{13}$$

The bound holds whatever the true source is. The prior of experts and users is used only to choose which tracers to buy, by the cost of the decision they change (T4.1). Monthly samples multiply the e-values, so the set can be read at every steering meeting and acted on at any time without losing the guarantee (Ville, 1939). A ledger shows what not knowing the source costs the utility, the farmers and the wetland. In the preparatory test the rule kept every source at or below 6.9% against a 10% budget, at €3.31 million. Pumping safe for all sources without tests cost €4.16 million, and €2.87 million if the source were known. Bayesian rules after the same tests reached 12.3% for the sources the prior did not favour. Monthly updates dropped the true source in 1.2–1.5% of cases, against 22% for an ordinary test repeated every month. A truth that mixes seawater and upconing breaks a list of pure sources (16.8%); admitting mixtures as hypotheses restores safety at a higher cost.

7. Gates, pass tests and fallbacks

Table 2. What each step must achieve before the next starts, and what happens if it does not.

Step and gate	Pass test	Fallback
Network design (WP4; MS1, Month 6)	Approved by the steering committee and the operators; certified optimality gap of at most 25%	Design re-run with the remaining budget; seismic nodes if fibre is not available
Network and coastal motion (WP4, WP1; MS2, Month 12)	At least 90% data availability; corrected heads within 3 cm root-mean-square of manual dips; vertical-velocity uncertainty of about 1–2 mm/yr at asset points	Corner reflectors and scatterers on structures where radar coherence is lost
Coupled model and unmeasured inputs (WP2; MS3, Month 18)	Held-out heads, chloride, InSAR and seismic velocity inside the 90% predictive intervals at close to the nominal rate; band for crossings at most 0.25 wide for each wellfield; the learned inland boundary passive	Model built in stages; two-dimensional sections through each asset with a wider error bound; a wider simulation-calibrated band, reported as such, if anchors are too few; a general-head boundary with a widened band if the inland boundary is not identified
Certified copy and alarm prototype (WP2, WP3; MS4, Month 22, go/no-go)	Error bound met on test runs; the prototype passes a dry run with the operators	Full model on a reduced domain for flagged forecasts
Back-testing (WP3; MS5, Month 27)	Each miss rate at or below its target with a 95% confidence bound, crossings between samples included, under the pre-registered protocol	Margin widened: the miss target is kept and the extra false alarms are reported
Live operation (WP5; MS6, Month 34)	Twelve months of logged bulletins; no e-process reaches 1/β; safe window reported for each asset; every pumping test below its chloride test limit; the plausible chloride sources and the pumping in force published every month	If no strong event occurs, verification rests on back-testing and storylines; smaller and shorter tests, or the window from routine records
Hand-over (WP6, WP7; MS7, Month 36)	An operating procedure adopted for each asset; the operators issue every bulletin of Months 34–36 without the project team; at least two trained people per operator; maintenance agreement signed by Month 33 (D6.2); open software and data with a DOI reproduced outside the team (D6.1, Month 33)	Extra training and tasks reassigned within the consortium; procedures restated jointly with the operator; the university hosts the system until the maintenance agreement is in place

8. What is new beyond the state of the art

The alarms assemble published methods of 2021–2026 (online and conditional conformal prediction, extreme conformal prediction, Learn then Test and anytime-valid verification); their joint use for coupled coastal hazards is an application and is not claimed as new. Eight contributions were derived and tested before the project (Table 3); the novelty statements rest on our literature search to date.

Table 3. The eight new contributions, the result of their preparatory test and where they are used.

Contribution	What is new	Preparatory test	Where
(i) Guarantees for quantities nobody measures	Bands for recharge, vadose delay, boundaries and dispersion that stay valid when the simulator or the rain product is wrong, with a computable floor	Raw ensemble missed 17–20% of weeks; calibrated bands 9.5–10% (target 10%)	T2.3; D2.1, MS3
(ii) The value of a sensor	The guaranteed band a sensor removes and the years of safe operation it adds	GNSS and corner reflectors: 5.6 and 5.0 years, where the classical criterion gives zero; a tide gauge rated useful widened the band by 16–21%	T4.1; D4.1, MS1
(iii) Pumping effects free of hidden confounding	The confounding path measured by the heads, with a calibrated check of every substitute	Bias 0.00 with the heads, against 0.56–0.59; check size 6%, power 99–100%	T5.3
(iv) Certified safe window and safe pumping tests	One window for three hazards and the year it closes; tests chosen for the years they add, with a chloride safety filter	Window closes after 38 years where single-hazard plans assume 43–97; tests: 10.8 to 31.9 years, chloride below 123 mg/L in 95% of runs	T5.3; D5.1, MS6
(v) A network that corrects its own sensors	Barometer and reference channels placed by design; robust choice with a certified optimality gap	+52% worst-case information; gap 18%; interval coverage 88–93% against 49–62%	T4.1–T4.3; D4.1
(vi) Warnings for what happens between samples	A certified band for a crossing at any time in the season, and the observation that closes it	Crossing 0.78 against 0.023 on a sampling date (34 times); band 0.18–1, narrowed to 0.76–0.80 by a logger	T3.2; D2.1, D3.2
(vii) Pumping before the source of the chloride is known	The sources that tracer e-values cannot exclude, kept without a prior; pumping safe for all of them; tracers bought for the decision they change; monthly updates valid at any time	Every source at or below 6.9% against a 10% budget (assuming seawater: up to 98%); €3.31 million against €4.16 million without tests; true source dropped in 1.2–1.5% of cases against 22%	T4.1, T5.3; MS6

Contribution	What is new	Preparatory test	Where
(viii) A boundary that learns the aquifer beyond the model	A conductance plus positive storage compartments: exact for any linear aquifer beyond the cut, passive whatever the rest of the model, learned from two piezometers while the pumping varies	Six compartments 7.8 mm against 148 mm for the best general-head boundary; unseen drought with doubled pumping 6 mm against 372 mm; fresh water lost to sea-level rise within 4% (prescribed inflow: 3.1 times too little)	T2.3, T4.1; D2.1, MS3

Together they answer questions that no current coastal warning answers for its operators: how often a warning will be missed and so what silence means; how many years of safe pumping and dewatering remain as the sea rises and the land sinks; which sensor or pumping test buys more of them; how the aquifer beyond the model answers new pumping and a rising sea; and how to pump safely before anyone knows where the salt comes from. The system starts at TRL 3, each component shown in preparatory tests, and ends at TRL 6, demonstrated on the operators' own sites through twelve months of live operation with the water utility, AMBU, the port authority, TAP and civil protection.

Software and data. The model, fast copies, alarms, safe window, test planner and network design are released under an open licence with containers and tests (D6.1). Operator data are used under the access agreements of T7.2 and the security rules of each operator; derived products, the bulletin log and the verification ledger are released openly following FAIR principles, and the preparatory tests are scripted and released with the software.

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TECHNICAL NOTE · THE ASSEMBLED SYSTEM · OCTOBER 2026

Assembling the equations of a coastal aquifer, with the placement of recharge and pumping wells as the final goal

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Faculty of Geology and Mining, Polytechnic University of Tirana, Albania. Assembles the equations of the author's series into one system for the Fushë-Kuqe aquifer, states the placement of recharge and pumping wells as its final goal, and adds the innovation MONTIMI-GO (montimi: the assembly).

Abstract. This note assembles into one system every equation the author's series has written for the Fushë-Kuqe coastal aquifer. The equations are: the flow equation with the two terms added to it (the memory of the silt and clay lenses, through a kernel κ_h and its tail K_h, and a learned closure Ψ); the salt equations with immobile zones; the long equation of seven ions with cation exchange, calcite, sulfate reduction and electro-diffusion; the toe and its mobility; the wells; and every boundary and initial condition. The assembled differential-algebraic system carries 31 unknowns per cell. Its block Jacobian couples flow and salt both ways and everything else one way, and its exact discrete adjoint gives the gradient of any goal in one backward sweep. The final goal is the placement of recharge and pumping wells. This is a chance-constrained, mixed-integer problem whose constraint is the chloride in the supply screen and whose safe yield is an order statistic of critical pumping rates. The innovation, MONTIMI-GO, lets this goal decide the assembly in four steps: one forward and one adjoint run weigh every term; the ion block is cut exactly, and terms below 2% of the limit are cut; the mesh is graded by the dual-weighted residual; the error that remains is calibrated into the constraint. On a vertical section with a 50-year pumping history the goal cut the memory of the lenses and the sea-level rise (weights at most 2.1×10^{-5}, a five-hundredth of the limit) and kept the immobile zones, dispersion, buoyancy and the same-step coupling (weights 3.0×10^{-3} to 2.6×10^{-2}); the one-pass split of flow and salt was 16% below the coupled screen chloride at half-year steps. The goal-guided assembly runs a 30-year forecast 2.5 times faster than the complete assembly, ranks the 25 layouts as it does (rank correlation 0.997) and, with its calibrated margin, chooses the supply well at 3.5 km and recharge well at 3.0 km at 5.95 m^2/d per metre of coast, whose risk judged on 25 draws of the complete assembly is 8% (2 of 25; 95% interval 1–26%), against a target of 10% and a guarantee of 20%, the target plus the calibration level. Run once afterwards, the ion block leaves the chloride goal unchanged and keeps sodium at the screen at 125 mg/L.

Keywords: seawater intrusion; variable-density flow; memory kernel; multi-species reactive transport; saltwater toe; equation assembly; discrete adjoint; goal-oriented error estimation; managed aquifer recharge; well placement; chance constraint; Fushë-Kuqe

1. What this note does

The series has written the equations of the Fushë-Kuqe coastal aquifer piece by piece: the variable-density flow of the base paper [1], the two terms added to the flow equation in reviews 42 and 43 and their synthesis, a memory of the silt and clay lenses and a learned closure [2–4], the long equation of multi-species transport with cation exchange, immobile zones and reactions that follows it [4], the salt equations and the chloride standard [5, 6], the mobility of the saltwater toe [7], the boundary and initial conditions [7–9], and the objectives that place monitoring, recharge and supply wells [1, 6, 10, 11]. Review 60 collected twenty papers from Oxford, Caltech, Harvard and MIT for the next step [12]. This note takes that step: it **assembles** all the equations into one system, and it states the placement of recharge and pumping wells as the **final goal** of the assembly.

Section 2 writes every equation in one notation, with the two added flow terms, the salt equations, the long multi-species equation, the toe and its mobility, the wells and every boundary and initial condition. Section 3 assembles them: the state, the discrete operators, the assembled differential-algebraic system, its block Jacobian and the order in which the blocks depend on one another. Section 4 states the final goal as a chance-constrained, mixed-integer problem constrained by the assembled system, and derives its adjoint. Section 5 is the innovation, MONTIMI-GO, an assembly set by the placement goal. Section 6 tests it on a vertical section of a coastal aquifer, solving the assembled system in full and in the assembly the goal chooses, and placing a recharge well and a supply well with both. Section 7 gives the recipe for Fushë-Kuqe and the limits, and Section 8 lists the system at a glance.

Table 1. Map of the note. The equation numbers refer to this note; the last column gives where each group was first written in the series.

Group of equations	Equations	Block of the assembled system	Source in the series
Flow with the two added terms (memory, closure), Darcy, density	(1)–(6)	flow block h, memory states h_k	base paper; reviews 42–43; synthesis
Salt (sea fraction) with immobile zones, dispersion, screen chloride	(7)–(12)	salt block c, immobile states c_k	base paper; chloride study; synthesis
The long equation: seven ions, exchange, reactions, electro-diffusion, invariants	(13)–(21)	ion block C, exchanger β	reviews 30, 42, 43; synthesis; leak note
Toe and its mobility	(22)–(25)	diagnostic X(t)	review 45 (KUFIRI-GO); this note
Wells	(26)–(28)	controls u	complete-system note; review 60
Boundary conditions	(29)–(32)	boundary data b(t)	reviews 44–45; assimilation note
Initial conditions	(33)–(35)	initial state y_0	reviews 44–45; assimilation note
Assembly	(36)–(43)	all	this note
Final goal: placing the wells	(47)–(53)	controls u, adjoint λ	complete-system note; review 60; this note
MONTIMI-GO	(54)–(57)	the assembly itself	this note

INNOVATION **MONTIMI-GO** (*montimi*: the assembly). The usual order is to assemble the most complete model one can and then optimise the wells on it. MONTIMI-GO reverses the order: the placement of recharge and pumping wells is stated first, and the adjoint of its constraint, the chloride at the supply screen, decides how the equations are assembled. One forward and one adjoint run of the complete system give the **weight** of every term for the decision. Blocks that the goal cannot see are cut exactly: the seven ions, the exchanger and the reactions (Theorem M1). Terms whose weight is below a small fraction of the chloride limit are cut, and their removal is predicted to first order. The mesh is graded by the dual-weighted residual of the goal. The error left by the cheap assembly is moved into the chance constraint as a calibrated margin, so that a layout accepted on the cheap assembly is accepted on the complete one. On the test of Section 6 the goal cut the memory of the lenses and the sea-level rise (weights at most 2.1×10^{-5}, a five-hundredth of the limit) and kept the immobile zones, dispersion, buoyancy and the same-step coupling (weights 3.0×10^{-3} to 2.6×10^{-2});; the goal-graded assembly runs a 30-year forecast 2.5 times faster than the complete assembly, ranks the 25 layouts as it does (rank correlation 0.997) and, with its calibrated margin, chooses the supply well at 3.5 km and recharge well at 3.0 km at 5.95 m^2/d per metre of coast, whose risk judged on 25 draws of the complete assembly is 8% (2 of 25; 95% interval 1–26%), against a target of 10% and a guarantee of 20%, the target plus the calibration level.

2. The equations, in one notation

The domain Ω is the aquifer between its top and its base, $x=(x, y, z)$ with z the elevation (m a.s.l.) and e_z the upward unit vector; time t is in days unless stated. The unknowns are listed in Table 2. Sea water is described by its fraction c, computed from chloride, the reaction invariant (Section 2.3), so that the density follows a conservative tracer.

Table 2. The unknowns of the assembled system.

Symbol	Unknown	Units	Equations
h_f	equivalent freshwater head	m	(1), (3)
h_k	head of the storage compartment k of the silt and clay lenses (Markovian lift of the memory)	m	(3)
q	Darcy flux	m/d	(4)
c	sea-water fraction from chloride	–	(7)
c_k	sea-water fraction in the immobile zone k	–	(8)
c	concentrations of Cl, Na, K, Ca, Mg, SO_4, HCO_3 in the mobile water	meq/L	(13)
c_k	ion concentrations in the immobile zone k	meq/L	(14)

Symbol	Unknown	Units	Equations
β	equivalent fractions of Na, K, Ca, Mg on the exchanger	–	(15)
X	position of the saltwater toe	m	(22)
u	controls: sites, recharge rates, pumping rates and screens of the wells	m^3/d	(47)
θ	uncertain parameters (conductivity, inflow, dispersivity, memory, sea level, ...)	various	(50)

2.1 Flow, with the two added terms

The classical equation of variable-density flow balances storage, the divergence of the Darcy flux and the wells [13, 14]. The series added two terms to it [2–4]: a **memory** of the silt and clay lenses, through a kernel κ_h and its tail K_h, and a **learned closure** Ψ for dynamics that are missing from the model:

$$S_s \partial_t h_f + \nabla \cdot q = \sum_w Q_w(t)\,\delta(x - x_w) - \int_{t_0}^{t} K_h(t-s)\,\partial_s h_f(x,s)\,ds + \Psi(h_f, x; \gamma) \tag{1}$$

where S_s is the specific storage (1/m), Q_w the rate of well w at x_w (positive for injection), K_h the memory tail and Ψ the closure with coefficients γ. The first added term is storage that is released late, by lenses that drain slowly; the second is everything the model leaves out, learned from data.

The memory was written in the synthesis article as a convolution of the head with a kernel κ_h. The two forms are the same, by the moment split (Lemma 1 of [4]):

$$\int_0^{\infty} \kappa_h(r)\,h_f(t-r)\,dr = \bar{\kappa}\,h_f(t) - \int_0^{\infty} K_h(r)\,\partial_t h_f(t-r)\,dr, \quad \bar{\kappa} = \int_0^{\infty} \kappa_h(r)\,dr, \quad K_h(r) = \int_r^{\infty} \kappa_h(u)\,du \tag{2}$$

where $\bar{\kappa}$ is the total weight of the kernel, an instantaneous leakage that belongs to the base condition (32), and K_h is the tail, a storage memory that acts only on changes of the head and vanishes at steady state. Equation (1) keeps the tail; the steady state of the assembled system is therefore the classical one.

For lenses that drain at rates a_k the tail is a sum of exponentials, completely monotone, and the memory has an exact Markovian lift with one head h_k per compartment, which is what is assembled:

$$K_h(r) = \sum_{k=1}^{K} s_k a_k e^{-a_k r} \Rightarrow \int_{t_0}^{t} K_h(t-s)\,\partial_s h_f\,ds = \sum_{k=1}^{K} s_k a_k (h_f - h_k), \quad \partial_t h_k = a_k (h_f - h_k) \tag{3}$$

where s_k (1/m) is the storage capacity of compartment k and a_k (1/d) its rate; the compartment heads start from their legacy $h_k(t_0)$, which absorbs the term $\sum_k s_k a_k e^{-a_k(t-t_0)}\left(h_f(t_0) - h_k(t_0)\right)$ of the integration by parts. In the kernel form $\kappa_h(r) = \sum_k s_k a_k^2 e^{-a_k r}$ and $\bar{\kappa} = \sum_k s_k a_k$ (Section 3.2 of [9]).

$$q = -K\left(\nabla h_f + \frac{\rho - \rho_f}{\rho_f} e_z\right), \quad K = diag(K, K, K_z), \quad h_f = \frac{p}{\rho_f g} + z \tag{4}$$

where K is the conductivity tensor for fresh water, p the pressure and g gravity. The buoyancy term makes salty water sink and drives the wedge.

$$\rho = \rho_f(1 + \delta c), \quad \delta = \frac{\rho_s - \rho_f}{\rho_f}, \quad c = \frac{Cl - Cl_f}{Cl_s - Cl_f} \tag{5}$$

where $\delta = 0.025$ is the density contrast and Cl_f, Cl_s the chloride of fresh water (12.92 mg/L, well 193) and of Adriatic sea water (21,781 mg/L) [6]. Viscosity is constant (Boussinesq).

$$\Psi(h_f, x; \gamma) = \sum_{j=1}^{J} \gamma_j \psi_j(h_f, x), J(\gamma) = \frac{1}{T} \int_0^T \left\| \dot{y} - f_0(y) - m(y; \gamma) \right\|^2 dt + R(\gamma) \tag{6}$$

where ψ_j are basis functions (zone indicators, h_f, $\left| \nabla h_f \right|$), f_0 the known model, m its learned correction and R a regulariser [9]. The closure is zero in every computed model of the series, including the test of Section 6; Section 6.2 gives where along the aquifer it would matter for the placement goal.

2.2 The salt equations

The sea-water fraction is transported with the flux of (4), dispersed by the Bear–Scheidegger tensor and exchanged with immobile zones in the silt and clay [4, 15]. In non-conservative form, which keeps c unchanged where water is released from storage:

$$n \partial_t c + \nabla \cdot (q c) - c \nabla \cdot q - \nabla \cdot (n D \nabla c) + \sum_{k=1}^{K} n b_k a_k (c - c_k) = \sum_w Q_w^+ \delta(x - x_w)(c_{inj,w} - c) \tag{7}$$

where n is the porosity, b_k the capacity ratio and a_k the rate of immobile zone k, Q_w^+ the injection rate of recharge well w and $c_{inj,w}$ the sea fraction of the water it injects (zero for fresh water). Pumping wells remove water at the resident concentration and need no term.

$$\partial_t c_k = a_k (c - c_k), k = 1, \dots, K \tag{8}$$

where c_k is the sea fraction in zone k; at Fushë-Kuqe zone 1 is the silt ($a_1 \approx 1$ yr^{-1}, $b_1 = 0.3$) and zone 2 the clay ($a_2 \approx 0.05$ yr^{-1}, $b_2 = 0.6$) [3].

$$n D = (n D_m + \alpha_T |q|) I + (\alpha_L - \alpha_T) \frac{q q^T}{|q|} \tag{9}$$

where D_m is the molecular diffusion and α_L, α_T the longitudinal and transverse dispersivities [16].

Eliminating the zones turns them into a memory of the salt, with its own kernel and a term for the initial disequilibrium (Proposition 1 of [4]):

$$n \left[\partial_t c + \int_{t_0}^{t} \kappa_c(s) \partial_t c(t - s) ds \right] + \nabla \cdot (q c) - c \nabla \cdot q - \nabla \cdot (n D \nabla c) = -n \sum_k b_k a_k e^{-a_k (t - t_0)} (c(t_0) - c \tag{10}$$

where $\kappa_c(s) = \sum_k b_k a_k e^{-a_k s}$; the source terms of (7) are omitted for brevity. The clay zone, with a time constant of twenty years, holds salt behind an advancing front and releases it behind a retreating one.

What a supply well delivers is the chloride averaged over its screen, which is the quantity the drinking-water standard is applied to [6, 17]:

$$Cl_w(t) = Cl_f + (Cl_s - Cl_f) \bar{c}_w(t), \bar{c}_w(t) = \frac{1}{L_w} \int_{z_w - L_w}^{z_w} c(x_w, z, t) dz \tag{11}$$

where x_w is the position of the well, z_w the top of its screen and L_w the screen length.

$$(Cl_w > 250\,\text{mg/L}) = (\bar{c}_w > \varepsilon_{lim}), \varepsilon_{lim} = \frac{250 - Cl_f}{Cl_s - Cl_f} = 0.01089 \tag{12}$$

where ε_{lim} is the sea fraction at which the screen reaches the chloride standard: about one per cent of sea water.

2.3 The long equation: seven ions, exchange and reactions

The equation that follows the flow in the synthesis article transports the seven major ions with an exchanger in local equilibrium, the immobile zones and the reactions, plus a learned closure confined to the span of the stoichiometric matrix [4]:

$$\partial_t\left(c+E\,s(c)\right)+\sum_{k=1}^{K} b_k\,a_k\left(c-c_k\right)+\nabla\cdot(v\,c)-\nabla\cdot(D\,\nabla c)=N\left(r(c)+\Phi(c,x,t;\vartheta)\right) \tag{13}$$

where $v=q/n$, E the exchange capacity (meq per litre of pore water, median 60), $s(c)$ the equivalent fractions on the exchanger for the cations and zero for the anions, N the stoichiometric matrix (Na–Ca, Na–Mg and K–Ca exchange, sulfate reduction, calcite), r the known rates and Φ the learned closure.

$$\partial_t c_k=a_k\left(c-c_k\right)+\Gamma_k c_k \tag{14}$$

where Γ_k holds the reactions inside zone k (sulfate reduction in the clay).

$$K_{Na\backslash I}=\frac{\beta_{Na}\,a_I^{1/z_I}}{\beta_I^{1/z_I}\,a_{Na}},\ I=K,Ca,Mg,\ \beta_{Na}+\beta_K+\beta_{Ca}+\beta_{Mg}=1,\ \beta_{Na}=\frac{2}{b_\beta+\sqrt{b_\beta^2+4\,a_\beta}} \tag{15}$$

where a_I are activities, z_I charges, the Gaines–Thomas selectivities are 0.2, 0.4 and 0.5, and $a_\beta=\left(c_{Ca}/K_{Ca}^2+c_{Mg}/K_{Mg}^2\right)/c_{Na}^2$, $b_\beta=1+c_K/\left(K_K\,c_{Na}\right)$ give the closed form [6, 18, 19].

$$\Omega=\frac{a_{Ca^{2+}}\,a_{CO_3^{2-}}}{K_{sp}},\ r_{cal}=\kappa\,g(\Omega),\ g(\Omega)=\begin{cases}\left(1-\Omega\right)^{n_1}, & \Omega\ge\Omega_c\\ \left(1-\Omega_c\right)^{n_1}\left(\dfrac{1-\Omega}{1-\Omega_c}\right)^{n_2}, & \Omega<\Omega_c\end{cases} \tag{16}$$

where Ω is the calcite saturation ratio ($\log K_{sp}=-8.43$ at 15 °C) and g a rate law with a near- and a far-from-equilibrium branch [6, 20].

$$\frac{dc_{SO_4,k}}{dt}=-k_{sr}\,c_{SO_4,k},\ \frac{dc_{HCO_3,k}}{dt}=+2\,k_{sr}\,c_{SO_4,k},\ \log_{10}\gamma_i=-0.509\,z_i^2\left(\frac{\sqrt{I}}{1+\sqrt{I}}-0.3\,I\right) \tag{17}$$

where k_{sr} is the rate of sulfate reduction in the clay (median 0.05 yr^{-1}), and γ_i the Davies activity coefficient with ionic strength I [21].

$$J_i=v\,c_i-\left(D+D_i\right)\nabla c_i-z_i D_i c_i\,\nabla\psi,\ \nabla\psi=-\frac{\sum_j z_j D_j\,\nabla c_j}{\sum_j z_j^2 D_j c_j},\ \sum_i z_i c_i=0 \tag{18}$$

where the electro-diffusive flux with zero electric current [22]; at Fushë-Kuqe the Péclet number is about 2.3×10^{5} and the leak of a reactive error into the chloride forecast is suppressed by about 4×10^{-6} (Theorem 2 of [23]).

$$u=W^{\top}c,\ W^{\top}N=0;\ f=\frac{Cl-Cl_f}{Cl_s-Cl_f},\ \Delta X=X_{obs}-\left[X_f+f\left(X_s-X_f\right)\right]=\left(N\,\xi\right)_X \tag{19}$$

$$BEX=Na+K+Mg-1.0716\,Cl \tag{20}$$

where the reaction invariants obey (13) without reactions; for these ions and reactions they are chloride and the charge balance, so chloride, hence c and the density, are free of every reaction and of the closure Φ (Proposition H1 of [6]). f, ΔX and BEX are the field indicators of mixing, reaction extents and base exchange [24].

$$\frac{dq_j}{dt}=\frac{1}{\tau_j}\left(\frac{Q_j f}{f+h_j}-q_j\right),\ \Phi_j=\frac{dq_j}{dt}\,min\left(1,\frac{c_{\iota(j)}}{c_{\iota(j)}^{mix}(f)}\right) \tag{21}$$

where the learned exchanger of GJURMA-GO, one example of the closure Φ: the loading q_j of exchanger j relaxes toward a learned equilibrium with the sea fraction f [2, 4].

2.4 The toe and its mobility

The toe is where the wedge of sea water meets the base of the aquifer. In the assembled system it is not a separate unknown but a functional of the salt field, the landward end of the half-sea-water line on the base:

$$X(t)=sup\left\{x:c\left(x,z_b,t\right)\geq\frac{1}{2}\right\} \tag{22}$$

where z_b is the elevation of the base. Where the base is not flat, c is taken on the base surface.

Its mobility follows from the salt equation by the implicit-function theorem: on the line $c=1/2$ the total change of c along the moving toe is zero, so

$$\frac{dX}{dt}=-\left.\frac{\partial_t c}{\partial_x c}\right|_{(X(t),z_b,t)} \tag{23}$$

where $\partial_t c$ is given by (7): the toe moves with the salt field, landward when the wedge gains salt at its tip and seaward when it loses it. This is exact and is what the assembled system computes.

Two reduced forms of the toe mobility are used for screening and for the boundary budget of review 45. The sharp-interface equilibrium of a confined aquifer with a seaward fresh flux q_s per metre of coast [16, 25], and a first-order relaxation toward it:

$$X_{eq}=\frac{K\delta B^2}{2q_s},q_s=q_{in}+\sum_r q_r-\sum_w Q_w,\frac{dX}{dt}=\frac{X_{eq}\left(u(t)\right)-X}{\tau_X} \tag{24}$$

where B is the thickness, q_{in} the regional fresh inflow and q_r, Q_w the recharge and pumping landward of the toe; τ_X is the time the wedge takes to adjust, lengthened by the immobile zones by up to the factor $1+\sum_k b_k$. KUFIRI-GO used the linearised form $dX/dt=\left(-\gamma h\left(0.5\,km\right)-X\right)/\tau_X$ with $\gamma=1,000$ m per m of head and $\tau_X=8$ yr [7]; Section 6.5 calibrates the relaxation law on the assembled system.

The toe is not what a supply well cares about. In the sharp-interface wedge the fresh water above the interface thickens landward as $B\sqrt{x/X}$, so a screen in the top L_w metres at x_w meets sea water only when

$$X\geq x_w\left(\frac{B}{L_w}\right)^2 \tag{25}$$

where which for the screens of Section 6 ($B=60$ m, $L_w=20$ m) asks the toe to lie nine times as far inland as the well. In the dispersive, pumped aquifer the mixing zone and the upconing below the well bring the standard much earlier; the toe is a late and indirect indicator of the event that matters, which is why the goal of Section 4 is the screen chloride (11) and not the toe.

2.5 Wells

$$\sum_w Q_w(t)\,\delta\left(x-x_w\right)\rightarrow\sum_w Q_w(t)\,\frac{\ell_w(z)}{L_w}\,\delta\left(x-x_w\right)\delta\left(y-y_w\right) \tag{26}$$

where each well is a line source over its screen, distributed by the length $\ell_w(z)$ open in each layer (by transmissivity where layers differ); supply wells have $Q_w<0$ over the top of the aquifer, recharge wells $Q_w=q_r>0$ over the full thickness.

$$h_{well}=h_{cell}+\frac{Q_w}{2\pi T}\ln\frac{r_e}{r_w},r_e\approx0.2\,\Delta x;\,c=c_{inj},c=c_{inj}\text{ at recharge wells} \tag{27}$$

where the head inside the well from the cell head (Peaceman [26]), needed when a drawdown limit is written for the well; injected water carries the composition of the recharge source (well 193) [6].

$$u=\left(z^R, z^S, q, Q, L\right), \sum_{k\in C_R} z_k^R q_k \rho_\epsilon\left(x-x_k\right), z_k^R \in \{0,1\} \to [0,1] \tag{28}$$

where z^R, z^S are the site indicators of recharge and supply wells over the candidate sets C_R, C_S, q and Q the rates, L the screens, and ρ_ϵ a smoothed source used when the indicators are relaxed to intensities for gradient-based placement [12].

2.6 Boundary conditions

The boundary is split into the sea face, the top (recharge, rivers, drains), the land margins and the base, with the wells inside [6–8]:

$$h_f=\frac{\rho_s}{\rho_f}\eta(t)-\delta z \text{ on } \Gamma_{sea}, \eta(t)=\eta_0+\dot{\eta}t+\sum_j A_j\cos\left(\omega_j t-\varphi_j\right)+\eta_{surge}(t)+\eta_{ia}(t) \tag{29}$$

where hydrostatic sea water in equivalent freshwater head; the sea level has a mean, a rise, tides, surges and interannual anomalies. In the areal models the sea is linked through a conductance (the Robin condition (32) of [6]).

$$c=1, c=c_{sea} \text{ where } q\cdot n<0, \left(n D\,\nabla c\right)\cdot n=0 \text{ where } q\cdot n\geq 0 \text{ on } \Gamma_{sea} \tag{30}$$

where the sea face switches with the direction of flow, for the salt and for every ion; n is the outward normal.

$$q\cdot n=-q_{in} \text{ or } 0, c=0, c=c_{193} \text{ on } \Gamma_{in}; n Z_r \frac{ds}{dt}=P-E(s)-L(s), -q\cdot n=R=L(s) \text{ on } \Gamma_{top} \tag{31}$$

where the land margins receive the regional fresh inflow or are closed; recharge is the drainage of a root-zone water balance with soil memory (moisture s, root depth Z_r, rainfall P, evaporation E) [7].

$$q_r=C_r\left(h_r(t)-max\left(h_f, z_b\right)\right), q_d=C_d\left(h_f-z_d\right)_+, -q\cdot n=\overline{\kappa}\left(h_f-h_{base}\right)+\int_{t_0}^{t} K_h(t-s)\,\partial_s h_f\, ds \text{ on } \Gamma_{bas} \tag{32}$$

where rivers and drains as Robin conditions, and a leaky base whose leakage follows the moment split (2): the instantaneous part $\overline{\kappa}$ and the memory tail.

2.7 Initial conditions

$$h_f(\cdot, t_0)=h_0, h_k(\cdot, t_0)=h_{k,0}, c(\cdot, t_0)=c_0, c_k(\cdot, t_0)=c_{k,0}, c(\cdot, t_0)=c_0, \beta(\cdot, t_0)=\beta_{eq}(c_0) \tag{33}$$

where every state of the assembled system needs an initial value; the memory states $h_{k,0}$, $c_{k,0}$ are the legacy of the lenses and the immobile zones.

$$h_0=\frac{\rho_s}{\rho_f}\eta(t_0)-\delta z \text{ on } \Gamma_{sea}, \sum_i z_i c_{0,i}=0, \sum_I \beta_{0,I}=1, \partial_t g(\cdot, t_0)=\left(f-A u_0\right)(\cdot, t_0) \text{ on } \partial\Omega \tag{34}$$

where compatibility of the boundary and initial data of order 0 and 1, charge balance and the closure of the exchanger [8].

$$y(t_0)=\Phi_{t_0\leftarrow t_{dev}}\left(y_{ss}; \text{pumping record}\right) \text{ or } p\left(y(t_0)/d_{0:T}\right)\propto p\left(y(t_0)\right)\prod_k p\left(d_k/y(t_k)\right) \tag{35}$$

where the present state is either the pre-development equilibrium y_{ss} carried forward through the pumping record by the assembled system itself (Φ is its flow map; this note), or the analysis of the data d by a smoother [9]. In either case the memory states are not in equilibrium at t_0.

3. The assembly

3.1 The state and the discrete operators

After discretisation in space by finite volumes on N cells, every field becomes a vector of cell values and every memory a set of local states. The assembled state is

$$y=\left(h,h_1,\ldots,h_K,c,c_1,\ldots,c_K,C,C_1,\ldots,C_K,\beta\right),\ dim\ y=N\left(2+2K+7\left(1+K\right)+4\right) \tag{36}$$

where h, c are the heads and sea fractions of the cells, h_k, c_k the memory states, C the $N\times 7$ array of ion concentrations and β the $N\times 4$ exchanger fractions. With $K=2$ there are 31 unknowns per cell; the Fushë-Kuqe grid of 5,589 active cells with 20 layers would carry about 3.5 million unknowns. The toe X is a functional of c, not a state.

On every face f between the cells f^- and f^+ (in the direction of increasing x or z), with area A_f, distance d_f and conductance $T_f=K_f\,A_f/d_f$, the flux and the salt flux are

$$Q_f=-T_f\left(h_{f^+}-h_{f^-}\right)-\left[f\ \text{vertical}\right]T_f\,d_f\,\delta\,\frac{c_{f^+}+c_{f^-}}{2},\ F_f=Q_f\,c_{\uparrow f}-\frac{\left(n\,D\right)_f\,A_f}{d_f}\left(c_{f^+}-c_{f^-}\right) \tag{37}$$

where $c_{\uparrow f}$ is the upwind value (the sea value 1 on a sea face with inflow) and $\left(n\,D\right)_f=n\,D_m+\left(\alpha_L\,q_n^2+\alpha_T\,q_t^2\right)/\left|q\right|$ the normal dispersion, with q_n the normal flux of the face and q_t the mean of the four perpendicular fluxes of its two cells. In matrix form $Q=G_h\,h+G_c\,c+g_0\left(t\right)$, where g_0 carries the sea level and the regional inflow; Div sums the outward fluxes of each cell, U selects the upwind cell and $Grad$ takes differences across faces.

3.2 The assembled system

Written block by block, with $M_S=diag\left(S_s\,V\right)$, $M_n=diag\left(n\,V\right)$, S_k, B_k the diagonal capacities and $A_k=a_k\,I$ the rates of the memory states, V the cell volumes and W, W^+ the well sources (net and injection), the semi-discrete system is

$$M_S\,\dot{h}=-Div\,Q\left(h,c,t\right)-\sum_k S_k\,A_k\left(h-h_k\right)+W\left(u\right)+V\,\Psi\left(h;\gamma\right),\ \dot{h}_k=A_k\left(h-h_k\right) \tag{38}$$

$$M_n\,\dot{c}=-Div\,F\left(Q,c\right)+diag\left(c\right)Div\,Q-\sum_k M_n\,B_k\,A_k\left(c-c_k\right)-diag\left(W^+\right)\left(c-c_{inj}\right) \tag{39}$$

where with $\dot{c}_k=A_k\left(c-c_k\right)$ for the immobile zones.

$$M_n\frac{d}{dt}\left(C+E\,S\left(\beta\right)\right)=-Div\,F\left(Q,C\right)+diag\left(Div\,Q\right)C-\sum_k M_n\,B_k\,A_k\left(C-C_k\right)+M_n\,N\left(r+\Phi\right) \tag{40}$$

where the third block is the long equation (13) with the same flux operator, $S\left(\beta\right)$ places the exchanger fractions on the cations, and the algebraic constraint $0=g_{ex}\left(C,\beta\right)$ is the Gaines–Thomas equilibrium (15) in every cell; the immobile ion states follow (14).

Collected, the assembly is a differential-algebraic system of index one (the exchanger fractions are fixed by the solution through the closed form of (15)):

$$M\,\dot{y}=F\left(y;\theta,b\left(t\right),u\left(t\right)\right),0=g\left(y\right),\ y\left(t_0\right)=y_0 \tag{41}$$

where θ are the uncertain parameters, $b\left(t\right)$ the boundary data of Section 2.6, $u\left(t\right)$ the controls of Section 2.5 and y_0 the present state (35).

3.3 Time stepping and the block Jacobian

Implicit Euler (or a Radau IIA method whose stage equations are generated from the same weak form [27]) turns (41) into one nonlinear system per step:

$$R_n\left(y_n,y_{n-1},z_{n-1};u_n\right)=M\,\frac{y_n-y_{n-1}}{\Delta t}-F\left(y_n\right)=0 \tag{42}$$

where z collects the local memory states, which are eliminated exactly in every cell:

$$h_k^n=\alpha_k\,h_k^{n-1}+\left(1-\alpha_k\right)h^n,\ \alpha_k=\frac{1}{1+a_k\,\Delta t}\Rightarrow S_k\,A_k\left(h^n-h_k^n\right)=S_k\,A_k\,\alpha_k\left(h^n-h_k^{n-1}\right) \tag{43}$$

where and likewise for c_k: the memory becomes a diagonal storage term and a source from the previous step, so it adds nothing to the size of the system that is solved.

The Jacobian of (42) has the block structure that decides how the system can be solved:

$$J_n=\frac{\partial R_n}{\partial y_n}=\begin{pmatrix} J_{hh} & J_{hc} & 0 & 0 \\ J_{ch} & J_{cc} & 0 & 0 \\ J_{Ch} & J_{Cc} & J_{CC} & J_{C\beta} \\ 0 & 0 & J_{\beta C} & J_{\beta\beta} \end{pmatrix} \tag{44}$$

where with the flow–salt block

$$J_{hh}=\frac{M_S}{\Delta t}+\sum_k S_k A_k \alpha_k + Div\, G_h,\; J_{hc}=Div\, G_c,\; J_{ch}=Div\, diag(c_\uparrow) G_h - diag(c)\, Div\, G_h \tag{45}$$

$$J_{cc}=\frac{M_n}{\Delta t}+\sum_k M_n B_k A_k \alpha_k + Div\left[diag(Q)U + diag(c_\uparrow)G_c - diag(d)\,Grad\right]$$

$$- diag(Div\, Q) - diag(c)\, Div\, G_c + diag(W^+) \tag{46}$$

where $c_\uparrow$ is Uc plus the sea inflow values, $d_f=(nD)_f A_f/d_f$. The dispersion coefficients are taken from the fluxes of the previous step, so the step also depends on y_{n-1} through them; that dependence enters the adjoint of Section 4.4. The test assembles these blocks exactly; Newton then converges quadratically (largest residuals 7×10^{-1}, 2×10^{-2}, 2×10^{-3}, 6×10^{-6}, 2×10^{-11}, 1×10^{-13} in one half-year step), which is the check that the blocks are right.

Proposition A1 (the order of the blocks). In the assembled system (38)–(40) the only two-way coupling between blocks is between flow and salt: the flux depends on the salt through the density, and the salt on the flow through the flux. The ion block, the exchanger, the reactions, the closure Φ and the toe are driven by (h, c) and feed nothing back, apart from electro-diffusion, which couples the ions back to chloride at a relative order of 4×10^{-6} at Fushë-Kuqe. The Jacobian (44) is therefore block lower-triangular below the 2×2 flow–salt block.

Proof. The density (5) depends on c, which is computed from chloride, and chloride is a reaction invariant (Proposition H1 of [6]), so neither the reactions nor Φ, which lies in the span of N, nor the exchanger enter the flow or the salt equations. The flux (4) depends on (h_f, c) only; the ion equations depend on Q and on the ions; the exchanger on the ions; the toe (22) on c. Hence $\partial R_h/\partial(C,\beta)=0$ and $\partial R_c/\partial(C,\beta)=0$. The electro-diffusive coupling (18) is the exception, bounded by Theorem 2 of [23]. □

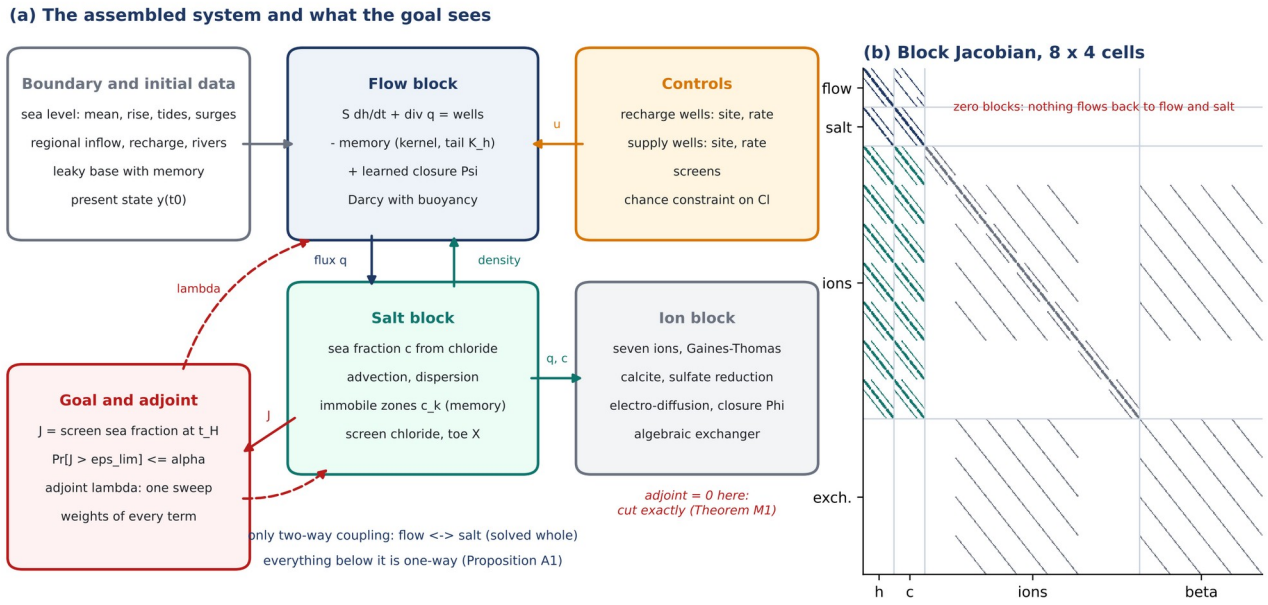


Figure 1. (a) The assembled system: boundary and initial data and the controls drive the flow block; flow and salt are coupled both ways (flux and density); the ion block is driven by them and feeds nothing back; the goal, the chloride at the supply screen, sends its adjoint back into the flow and salt blocks only. (b) The pattern of the assembled Jacobian (44) on a grid of 8 × 4 cells: flow–salt block (dark), ion and exchanger blocks (grey) below it, and the empty blocks above them that make the system block lower-triangular (Proposition A1).

3.4 Two clocks, and why the flow–salt block is solved whole

Heads diffuse with $D_h = T/S \approx 2 \times 10^7$ m²/d and feel a pumping change kilometres away within a day; salt moves kilometres in decades. The ratio of the two clocks is about 10^{-5} [4], which suggests solving the flow and the salt one after the other, with the density lagged by a step. The test shows that this split, though consistent, is not harmless for the goal. Picard iteration between the two blocks did not converge at half-year steps (the change of the sea fraction between successive iterations grew from 6×10^{-2} after the first to 3×10^{-1} after thirty), while Newton on the whole block converged quadratically. The one-pass split approached the coupled value of the screen sea fraction only as the step was cut: it was 24%, 16%, 10%, 2% below the coupled value at steps of 1, 0.5, 0.25, 0.125 years, while the coupled value itself changed by at most 0.7% over the same steps. The flow–salt block is therefore solved whole, as in the monolithic solvers of [28, 29], and Section 5 lets the goal confirm that choice.

3.5 The reductions of the series as cuts of the assembly

Table 3. The models computed in the series are cuts of the assembled system. MONTIMI-GO chooses the cut by the goal instead of by habit.

Model of the series	Blocks kept	Blocks cut or frozen	Exact for
R1: areal steady flow with the Ghyben–Herzberg test [1]	flow (steady, depth-integrated)	salt, memory, ions; interface from the head	head-based questions at steady pumping
R2: Strack potential with a Gaussian mixing zone [5]	flow (steady), sharp interface + mixing width	salt transport, memory, ions	steady screen chloride, threshold form of P1–P2 [6]
R3: vertical variable-density section (this test, Section 6)	flow, salt, memory, immobile zones, ions	the third dimension	transients of a cross-section
R4: seven-ion flow line [2, 3]	salt, ions, exchange, memory	flow (prescribed velocity history)	chemistry along a flow path
Toe relaxation law (24) [7]	one ODE for X	everything else	the toe, slowly varying forcing

4. The final goal: placing recharge and pumping wells

4.1 Decisions, goal and constraint

The decision is where to put recharge wells and supply wells, how much to inject and pump and how to screen, within the constraints of the series (spacing, exclusion around supply wells, budget; (50) of [6]):

$$u=\left(z^{R},z^{S},q,Q,L\right)\in U,U=\left\{\left|x_{k}-x_{l}\right|\geq d_{min},\left|x_{r}-x_{w}\right|\geq d_{ex},\sum_{j}c_{j}z_{j}\leq B\right\} \tag{47}$$

where z^{R}, z^{S} are binary site indicators over the candidate sets, q, Q the recharge and pumping rates and L the screens; c_{j} are costs and B the budget.

The goal of every supply well is the screen chloride (11) at the end of the planning horizon t_{H}; following PUSI-GO [6], chloride is the standard that binds, and sodium is checked afterwards (Proposition P3 there):

$$J_{w}\left(u;\theta\right)=\bar{c}_{w}\left(t_{H};u,\theta\right)=w_{w}^{\top}c\left(t_{H}\right),\text{event}\max_{w}J_{w}>\varepsilon_{lim} \tag{48}$$

where w_{w} averages the cell values over the screen of well w (interpolated linearly to x_{w}, so that every grid computes the same functional).

$$\max_{u\in U}\sum_{w}v\left(x_{w}\right)Q_{w}-\lambda_{R}\sum_{r}q_{r}\,\text{s.t.}\,\Pr_{\theta}\left[\max_{w}J_{w}\left(u;\theta\right)>\varepsilon_{lim}\right]\leq\alpha,M\,\dot{y}=F\left(y;\theta,b,u\right),0=g\left(y\right) \tag{49}$$

where $v\left(x_{w}\right)$ is the value of a cubic metre pumped at x_{w}, net of conveyance to the demand at the coast, and λ_{R} the price of recharged water. The problem is mixed-integer (the sites), chance-constrained (the posterior of θ) and constrained by the assembled system [30–32]. At scale it can be solved with learned constraints inside a mixed-integer program [33], learned neighbourhood search [34] or Bayesian optimisation over mixed spaces [35], with the sites fixed first and the rates as recourse [36]; with a few candidate sites, as in the test, it is enumerated.

$$\frac{1}{N_{\theta}}\sum_{i=1}^{N_{\theta}}1\left\{\max_{w}J_{w}\left(u;\theta_{i}\right)>\varepsilon_{lim}\right\}\leq\alpha \tag{50}$$

where the sample-average form over N_{θ} posterior draws, which is what is solved.

Proposition G1 (the safe yield of a layout is a quantile). Fix the sites and the recharge of a layout ℓ and let $J\left(\ell,Q;\theta\right)$ be nondecreasing in the pumping Q for every θ. Define the critical rate $Q_{c}\left(\ell;\theta\right)=sup\left\{Q\geq0:J\left(\ell,Q;\theta\right)\leq\varepsilon_{lim}\right\}$. Then the largest Q that satisfies (50) is the order statistic

$$Q^{\star}\left(\ell\right)=Q_{c,(m)}\left(\ell\right),m=\lfloor\alpha N_{\theta}\rfloor+1 \tag{51}$$

where the m-th smallest of the critical rates of the draws, and the placement problem becomes a choice among layouts of the value $v\,Q^{\star}\left(\ell\right)-\lambda_{R}q\left(\ell\right)$.

Proof. By monotonicity $J\left(\ell,Q;\theta_{i}\right)>\varepsilon_{lim}$ if and only if $Q>Q_{c}\left(\ell;\theta_{i}\right)$, so the left side of (50) is the fraction of draws with $Q_{c}<Q$; it is at most α exactly for $Q\leq Q_{c,(m)}$. □

Monotonicity holds in every run of the test (more pumping never freshened a screen within the horizon). It reduces the chance constraint to one root-finding per draw and layout, each a few forward runs of the assembled system.

Remark G2 (on the draws and beyond them). Constraint (50) holds on the draws used. For a new draw exchangeable with them, and without ties, the probability that its critical rate falls below Q_c,(m) is m/(N_θ + 1): 3/26 = 11.5% for N_θ = 25 and α = 10%, slightly above α. Taking m = ⌊α(N_θ + 1)⌋ instead keeps it at most α (2/26 = 7.7% here). A statement that holds with high probability, rather than on average over the draws, needs a binomial bound and more draws.

4.2 The adjoint of the assembled system

Gradients of the goal with respect to every rate and every candidate site, and the weights of Section 5, come from one backward sweep. With λ_{n} the adjoint of step n and μ_{n} that of the memory states,

$$J_{n}^{\top}\lambda_{n}=\delta_{n,N_{t}}\frac{\partial J}{\partial y_{N_{t}}}^{\top}-\left(\frac{\partial R_{n+1}}{\partial y_{n}}\right)^{\top}\lambda_{n+1}+B^{\top}\mu_{n},\mu_{n-1}=\alpha\,\mu_{n}-\left(\frac{\partial R_{n}}{\partial z_{n-1}}\right)^{\top}\lambda_{n} \tag{52}$$

where for $n = N_t, \dots, 1$ with $\lambda_{N_t+1} = 0$, $\mu_{N_t} = 0$; B and α are the diagonal maps of (43); a last solve with the Jacobian of the pre-development equilibrium carries the adjoint into the initial state. $\partial R_{n+1} / \partial y_n$ contains the storage and the dependence of the dispersion coefficients on the previous fluxes.

$$\frac{dJ}{dp} = -\sum_n \lambda_n^{\top} \frac{\partial R_n}{\partial p}, \quad \frac{dJ}{dQ_w} = -\sum_n \lambda_{h,n}^{\top} e_w, \quad \frac{dJ}{dq_r(x)} = \sum_n \left(\lambda_{h,n} - diag(c_n)\lambda_{c,n}\right)^{\top} e_r(x) \tag{53}$$

where e_w, $e_r(x)$ distribute a unit pumping over the supply screen and a unit injection over the full thickness at the candidate x; one sweep gives the sensitivity of the goal to a recharge well at every column of the grid. In the test the adjoint gradients agree with central differences of the complete assembly to 8 significant digits for the pumping rate and for the recharge rate (supply well at 2.0 km, recharge well at 1.0 km, step 10^{-5}). At a layout where the net seaward flux past the supply well nearly vanishes they agree to 0.3%: there some face fluxes change direction under the perturbation, and the upwind goal is only piecewise differentiable.

5. MONTIMI-GO: an assembly set by the placement goal

The assembled system of Section 3 is large, and the placement problem of Section 4 asks for it thousands of times: one root-finding per layout and posterior draw, each a few forecasts. Review 60 proposed to let the goal decide the assembly [12]. Four results make this concrete. The first is exact, the second and third are first-order, and the fourth moves what is left into the constraint.

Theorem M1 (exact cuts). Let the goal J depend only on the flow and salt states (h, c), as the screen chloride (48) does, and let the assembled Jacobians be block lower-triangular below the flow–salt block (Proposition A1). Then at every step the adjoint vanishes on every downstream block, $\lambda_{C,n} = 0$ and $\lambda_{\beta,n} = 0$. Consequently (i) J and its gradient with respect to every control are the same whether or not the ions, the exchanger, the reactions and the closure Φ are assembled; (ii) the weight (54) of every parameter inside those blocks (selectivities, exchange capacity, reaction rates, closure coefficients) is zero; (iii) with electro-diffusion, (i) and (ii) hold up to the relative order of the leak bound, 4×10^{-6} at Fushë-Kuqe [23].

Proof. In the adjoint recursion (52) the right-hand side has no downstream component at $n = N_t$, because J does not depend on (C, β). The transposed Jacobian $J_n^{\top}$ is block upper-triangular, so its downstream rows read $J_{CC}^{\top}\lambda_C + J_{\beta C}^{\top}\lambda_\beta = r_C$ and $J_{C\beta}^{\top}\lambda_C + J_{\beta\beta}^{\top}\lambda_\beta = r_\beta$, with an invertible diagonal block (the ion transport operator and the exchange equilibrium of index one). With $r_C = r_\beta = 0$ the only solution is zero. The cross-step Jacobian $\partial R_{n+1} / \partial y_n$ has the same triangular structure, so the downstream right-hand side stays zero at every earlier step, by induction backwards. Statement (i) follows because the flow–salt rows of the forward system do not contain the downstream unknowns. Statement (ii) follows because a parameter p of a downstream block has $\partial R_n / \partial p$ supported on downstream rows only, so $\lambda_n^{\top} \partial R_n / \partial p = 0$. Statement (iii) follows from the perturbation bound of [23]. □

The theorem is the reason the series could put a learned chemistry closure into the model without touching the chloride forecast (the "cut" of [4, 23]), here turned into a rule of assembly: for a chloride goal the ion block is computed once, after the decision, to check sodium and hardness for the chosen layout (Proposition P3 of [6]).

Proposition M2 (the goal weight of a term). Give every term T of the assembly a scale s_T, with $s_T = 1$ for the term as assembled and $s_T = 0$ for the term removed; for the coupling, let s_T interpolate between the same-step density and the density of the previous step. The weight of T for the goal, and the first-order change of the goal when T is removed, are

$$w_T = \left.\frac{dJ}{ds_T}\right|_{s_T = 1} = -\sum_n \lambda_n^{\top} \frac{\partial R_n}{\partial s_T}, \quad J|_{s_T = 0} - J|_{s_T = 1} = -w_T + O\left(\|\Delta y\|^2\right) \tag{54}$$

where with one forward and one adjoint run of the complete assembly for all terms at once. The rule of assembly is

$$\text{cut } T \;\Leftrightarrow\; \max_{\text{reference cases}} |w_T| < \tau\, \varepsilon_{lim} \tag{55}$$

where with τ a small fraction (here 2%, i.e. $|w_T| < 2.2 \times 10^{-4}$, about 5 mg/L of chloride at the screen). The weight is the derivative with respect to the scale of the term, not the residual difference at a fixed state: removing the sea-level rise at a fixed state shifts the heads of the sea face by tens of centimetres and reverses inflow and outflow there, while the actual change of the goal is nil because the whole head field rises with the sea.

Proof. J is a function of s_T through the solution of $R_n(y; s_T) = 0$; differentiating the constraints and using (52) gives $dJ/ds_T = -\sum_n \lambda_n^T \partial R_n / \partial s_T$ [37, 38]. Taylor's theorem in s_T gives the remainder. □

Proposition M3 (a mesh graded by the goal). Let y_H be the solution of the assembly on a coarse mesh and $I\, y_H$ its interpolation to the complete mesh h. The dual-weighted residual estimates the error of the goal and gives a cell-wise indicator [39]:

$$J_h(I\, y_H) - J_h(y_h) \approx \sum_n \lambda_n^T R_n(I\, y_H), \eta_j = \sum_n \sum_{i \in column\, j} \left| \lambda_{n,i} R_{n,i}(I\, y_H) \right| \tag{56}$$

where the column indicator η_j is summed over the reference cases. The goal-graded mesh equidistributes $(\eta_j / \Delta x_j)^{1/3}$ over the same number of columns as the uniform coarse mesh, with a cap on the widest column.

The estimate itself is only first order and, for a goal that sits in the tail of a sharp front, it was not reliable in the test (Section 6.3); learned conservative fluxes are one way to keep such fronts sharp on coarse cells [40]. MONTIMI-GO therefore uses the indicator only to rank where the mesh matters, and does not rely on the estimate; the remaining error is calibrated by Proposition M4.

Proposition M4 (the error of the cheap assembly moves into the constraint). Let A be the cheap assembly and F the complete one. Draw C calibration cases (ℓ_i, θ_i) like the decision cases, compute $r_i = Q_c^A(\ell_i; \theta_i) - Q_c^F(\ell_i; \theta_i)$, and let m_β be the $\lceil (1-\beta)(C+1) \rceil$-th smallest r_i. Solve the placement on A with the critical rates lowered by m_β:

$$Q^\star_{A,\beta}(\ell) = \left[Q^A_{c,(m)}(\ell) - m_\beta \right]_+, \quad Pr\left[J_F\left(\ell, Q^\star_{A,\beta}(\ell); \theta\right) > \varepsilon_{lim} \right] \le \alpha + \beta \tag{57}$$

where for a new draw θ exchangeable with the calibration cases, up to the sampling error of the draws.

Proof. If the case is exchangeable with the calibration cases, $Pr[r > m_\beta] \le \beta$ by the rank argument of split conformal prediction [41, 42]. On $\{r \le m_\beta\}$ we have $Q_c^F \ge Q_c^A - m_\beta$, so exceeding the standard on F at $Q^\star_{A,\beta}$ implies $Q_c^A - m_\beta < Q^\star_{A,\beta}$, i.e. $Q_c^A < Q^A_{c,(m)}$, which has probability at most α (Proposition G1). A union bound gives $\alpha + \beta$. □

The procedure. (1) Choose a few reference cases: layouts at their critical rates on the complete assembly. (2) For each, one forward and one adjoint run of the complete assembly give the weights of all terms, the gradients with respect to every rate and every recharge site, and the mesh indicator. (3) Cut the downstream blocks (Theorem M1) and the terms below the tolerance; keep the rest; grade the mesh. (4) Run C calibration cases on both assemblies and compute the margin. (5) Solve the placement on the cheap assembly with the margin. (6) Run the complete assembly, ions included, once for the chosen layout: chloride against the margin, sodium and hardness against Proposition P3 of [6]. The cheap assembly can be replaced or completed by a neural-operator surrogate trained on runs of the assembly [43, 44], or differentiated end to end for gradient-based placement [45]; the margin of Proposition M4 applies to either, and the decision should then be operated and revalidated as in field practice for injection wells [46].

6. The test

6.1 The aquifer, the history and the decision

The test assembles the system on a vertical section of a confined coastal aquifer, the reduction R3 of Table 3. It is 4 km long and 60 m thick, its top is 10 m below sea level, and it has silt and clay lenses with hydraulic memory, immobile zones, and the ion block with exchange. The parameters are those of Fushë-Kuqe where the series has them (Table 4). The present state is made as at the field site, where the coastal screens salinised under pumping

[47]: from the pre-development equilibrium, 50 years of pumping 4 m^2/d per metre of coast from a well field 1.5 km from the coast. At the end of the history its screen delivers 364 mg/L of chloride in the median aquifer and the toe has moved from 349 m to 1070 m. The decision is where to put the new supply well (2.0, 2.5, 3.0 or 3.5 km from the coast), where to put one recharge well injecting 1 m^2/d of fresh water (none, 0.5, 1.0, 1.5 km, the old well field, 2.0, 2.5 or 3.0 km), and how much to pump, so that the probability that the supply screen exceeds 250 mg/L at the end of a 30-year horizon is at most 10%: 25 layouts.

Table 4. The test bed. ξ denotes independent standard normal variables.

Quantity	Value or prior	Source
Grid of the complete assembly	120 × 20 cells (33 m × 3 m); half-year steps	this note
Conductivity K, anisotropy, porosity	$100\,e^{0.25\,\xi}$ m/d (T ≈ 4,700–7,700 m^2/d), $K_z/K = 0.1$, $n = 0.25$	complete-system note, Table 2
Specific storage; lens capacities and rates	5×10^{-6} 1/m; $s_k = 2 \times 10^{-5}, 5 \times 10^{-5}$ 1/m; $a_k = 1$, 0.05 yr^{-1}	reviews 43–45 (rates); this note
Immobile zones	$b_1 = 0.3$, $b_2 \sim U(0.3, 0.9)$; $a_1 = 1$, $a_2 = 0.05$ yr^{-1}	RITMI-GO
Dispersivities	$\alpha_L = 20\,e^{0.4\,\xi}$ m, $\alpha_T = \alpha_L/10$	chloride study (10 m, 1 m) widened
Regional fresh inflow	$4\,e^{0.2\,\xi}$ m^2/d per metre of coast	recharge 0.5 mm/d over about 8 km
Sea-level rise	$N(4, 2^2)$ mm/yr	KUFIRI-GO
Density contrast; end-members; standards	0.025; well 193 and Adriatic water; Cl 250, Na 200 mg/L	complete-system note
Exchange	E = 60 meq/L; selectivities 0.2, 0.4, 0.5	GJURMA-GO
Draws	25 design draws (the cheap assemblies choose; 10 for the two counter-examples), 25 evaluation draws (the complete assembly judges) and 40 calibration cases for the margin (16 for the counter-examples)	this note

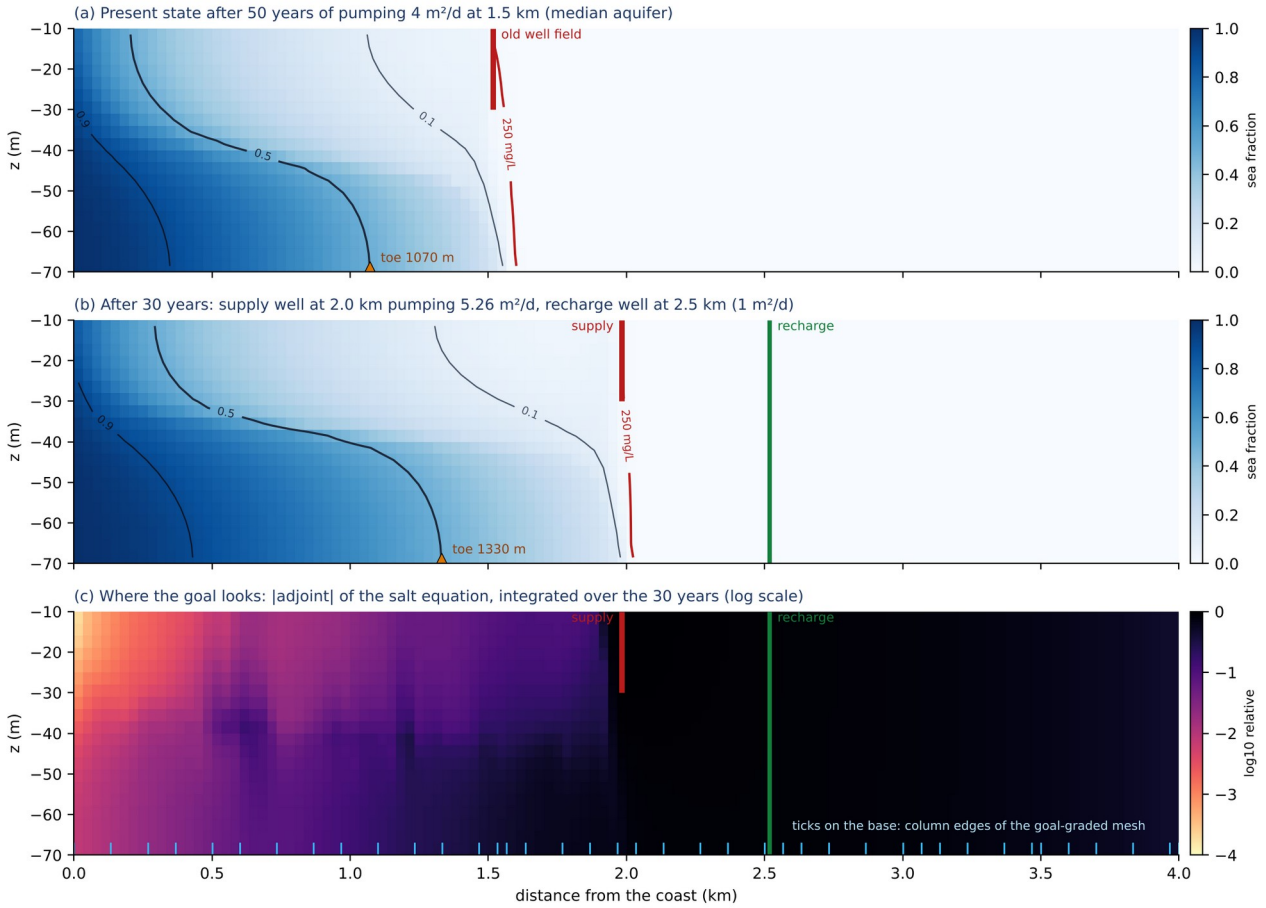


Figure 2. The test bed with the median parameters. (a) The present state after the 50-year history: the old screen at 1.5 km delivers 364 mg/L of chloride; the toe (triangle) is at 1070 m, against 349 m before development. (b) Thirty years later with a supply well at 2.0 km pumping 5.26 m^2/d per metre of coast, its critical rate, and a recharge well at 2.5 km; the red line is the 250 mg/L contour, which touches the bottom of the screen; the toe has moved to 1330 m. (c) Where the goal looks: the adjoint of the salt equation for the screen chloride, integrated over the 30 years. The ticks on the base are the column edges of the goal-graded mesh.

6.2 What the goal kept and what it cut

Table 5 and Figure 3 give the weights of the eight terms at four reference cases, each a layout at its critical rate on the complete assembly, so that the screen sits exactly at 250 mg/L: S2.0-R2.5, S2.5, S3.0-R1.5, S3.5-R1.0 (S: supply well, R: recharge well, km from the coast). One forward and one adjoint run of the complete assembly took 11 s and about 4 s per case. The hydraulic memory of the lenses and the sea-level rise weigh at most 2.1×10^{-5}, a five-hundredth of the limit. The lenses release stored water slowly, but at these capacities they change the head at the screen by millimetres. The rise lifts the whole head field with the sea and moves almost no water. The rule cut all three; removing them changed the goal by at most 2.1×10^{-5}, as predicted to within 25%, and to within 1% in three of the four cases. The immobile zones, transverse dispersion and buoyancy weigh 6.3×10^{-3} to 2.6×10^{-2}, between half and twice the limit itself: they set the width and the speed of the mixing zone that reaches the screen. A 10% reduction of the immobile capacities or of the transverse dispersivity was predicted to within 8%. The same-step coupling weighs 3.0×10^{-3} to 6.6×10^{-3} and was kept. For this term the first-order number only says that it matters: the goal is not monotone in the lag (it rose by 5.5×10^{-5} for 1% of lag and by 7.5×10^{-4} for 20% in a separate check), and the full one-pass split moved it by -4.3×10^{-3} to -1.9×10^{-3}, in the opposite direction to the first-order value.

Table 5. Goal weights w_T of (54) over the four reference cases (sea fraction at the screen per unit scale of the term; the limit is $\varepsilon_{lim}=0.0109$ and the tolerance $\tau\,\varepsilon_{lim}=2.2\times10^{-4}$). Check: the change of the goal predicted from the weight against the change computed by running the complete assembly without the term (cut terms), with the term reduced by 10% (kept terms), or with the one-pass split.

Term	Weight w_T (range over 4 cases)	Check: predicted / actual change	Decision
Hydraulic memory, silt lenses	6.6×10^{-7} – 9.6×10^{-6}	removed: 9.6×10^{-6} / 9.6×10^{-6} (largest)	cut
Hydraulic memory, clay lenses	7.8×10^{-6} – 2.1×10^{-5}	removed: 2.1×10^{-5} / 2.1×10^{-5} (largest)	cut
Immobile zone, silt	6.3×10^{-3} – 1.4×10^{-2}	−10%: 6.3×10^{-4} / 6.2×10^{-4} (first case)	kept

Term	Weight w_T (range over 4 cases)	Check: predicted / actual change	Decision
Immobile zone, clay	1.0×10^{-2} – 1.2×10^{-2}	−10%: 1.1×10^{-3} / 1.1×10^{-3} (first case)	kept
Transverse dispersion	7.5×10^{-3} – 1.3×10^{-2}	−10%: 1.3×10^{-3} / 1.4×10^{-3} (first case)	kept
Sea-level rise	3.6×10^{-6} – 6.8×10^{-6}	removed: 6.8×10^{-6} / 7.7×10^{-6} (largest)	cut
Buoyancy (density coupling)	9.8×10^{-3} – 2.6×10^{-2}	δ −10%: -2.6×10^{-3} / -3.8×10^{-3} (first case; δ also enters the sea face)	kept
Same-step flow–salt coupling	3.0×10^{-3} – 6.6×10^{-3}	one-pass split: -3.0×10^{-3}… / -4.3×10^{-3} to -1.9×10^{-3}	kept

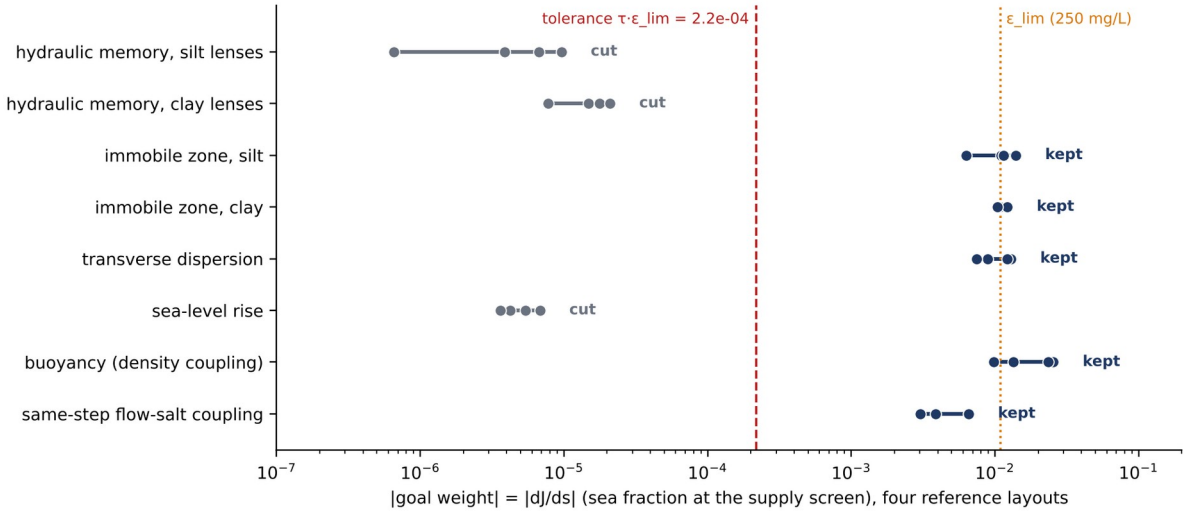


Figure 3. The goal weights of every term of the assembly, four reference layouts (dots). Grey terms lie three orders of magnitude below the tolerance (dashed) and are cut; dark terms lie near the limit itself (dotted) and are kept. The ion block is not shown: its weight is zero by Theorem M1.

The learned closure Ψ is zero in the test, but its weight shows where dynamics missing from the model would matter for the decision. A missing source equal to 10 mm/yr of recharge spread over a 500-m band changes the screen sea fraction by at most 2.8×10^{-5} in the band at the coast and by up to 7.6×10^{-4} (7% of the limit) in the bands landward of the supply well, where every litre of fresh water passes the well. The closure, and the data that identify it, are therefore worth most for the inland water balance, not at the coast.

6.3 The mesh

The indicator (56), summed over the four reference cases, is largest where the supply screens draw water through the mixing zone (Figure 2c). The goal-graded mesh puts columns of 33–67 m at the candidate supply sites and at the old well field and columns of 100–133 m elsewhere (39 columns; the uniform coarse mesh has 40 columns of 100 m). Table 6 compares, on the four reference cases and eight further cases with random layouts and draws, the error of each coarse assembly in the goal with its dual-weighted estimate. Both coarse meshes overestimate the screen sea fraction, because a coarse cell smears the mixing zone upward. The goal-graded mesh does so on every case by 6.0×10^{-3} to 1.3×10^{-2}, and its estimate has the right sign and order on every case (ratio of estimate to error 0.40–0.87). The uniform mesh errs by anywhere between 2.4×10^{-3} and 1.8×10^{-2}, depending on where the supply well falls relative to its columns, and its estimate has the wrong sign in 6 of 12 cases. The goal-graded mesh is therefore not always the more accurate, but its error is the more **predictable**, which is what Proposition M4 needs.

Table 6. Error of the coarse assemblies in the goal (screen sea fraction, limit 0.0109) at the critical rate of the complete assembly, against the dual-weighted-residual estimate (56). Uniform: 40 columns of 100 m; goal-graded: 39 columns of 33–133 m. Same physics as the complete assembly.

Case	True error, uniform	Estimate, uniform	True error, goal-graded	Estimate, goal-graded
reference: S2.0-R2.5, Q = 5.26	2.36×10^{-3}	-2.02×10^{-2}	1.28×10^{-2}	5.32×10^{-3}
reference: S2.5, Q = 5.12	1.53×10^{-2}	8.45×10^{-3}	1.05×10^{-2}	6.08×10^{-3}
reference: S3.0-R1.5, Q = 6.20	1.21×10^{-2}	1.06×10^{-2}	9.04×10^{-3}	7.84×10^{-3}

Case	True error, uniform	Estimate, uniform	True error, goal-graded	Estimate, goal-graded
reference: S3.5-R1.0, Q = 6.74	2.69×10^{-3}	-4.65×10^{-3}	5.98×10^{-3}	4.63×10^{-3}
random: S3.5-R2.0, Q = 10.28	3.99×10^{-3}	-2.42×10^{-3}	8.43×10^{-3}	6.22×10^{-3}
random: S3.5-R1.0, Q = 9.02	2.77×10^{-3}	-3.93×10^{-3}	7.39×10^{-3}	5.76×10^{-3}
random: S2.0-R3.0, Q = 3.84	3.47×10^{-3}	-1.67×10^{-2}	1.11×10^{-2}	4.48×10^{-3}
random: S3.5, Q = 6.62	3.49×10^{-3}	-4.51×10^{-3}	7.48×10^{-3}	5.85×10^{-3}
random: S3.0, Q = 4.46	1.20×10^{-2}	1.11×10^{-2}	8.11×10^{-3}	6.95×10^{-3}
random: S2.5-R3.0, Q = 7.21	1.81×10^{-2}	8.09×10^{-3}	1.28×10^{-2}	7.18×10^{-3}
random: S3.0-R0.5, Q = 3.95	1.13×10^{-2}	9.92×10^{-3}	7.03×10^{-3}	5.23×10^{-3}
random: S3.0-R2.5, Q = 6.94	1.04×10^{-2}	9.88×10^{-3}	8.16×10^{-3}	6.23×10^{-3}

6.4 Placing the wells

Every assembly solves the placement problem (49) the same way. For each of the 25 layouts and each draw it finds the critical rate by a bracketed secant on the logarithm of the goal (Proposition G1; 5.5 forecasts per root on average). It takes the safe yield as the order statistic (51) at $\alpha = 10\,\%$, and chooses the layout of largest value $v(x_s)Q^{\star} - \lambda_R q_r$. The value counts water at the supply well less 0.2 per km of conveyance beyond 2 km, and recharged water at 0.5 of its value. The complete assembly does this on 25 evaluation draws and acts as the referee. The cheap assemblies do it on separate design draws (25 for the goal-guided and uniform assemblies, 10 for the two counter-examples), with and without the calibrated margin of Proposition M4 estimated from 40 calibration cases (16 for the two counter-examples). The full assembly needed 15 minutes per draw for all layouts.

Table 7. The placement on each assembly. Layout chosen by the value $v(x_s)Q^{\star} - \lambda_R q_r$ with $v = 1 - 0.2(x_s - 2\,km)/km$ and $\lambda_R = 0.5$; safe rate in m² per day per metre of coast; risk: fraction of the 25 evaluation draws of the complete assembly on which the screen exceeds 250 mg/L at that layout and rate (target 10%; 1 of 25 draws is 4%, 95% interval 0.1–20%, and 2 of 25 is 8%, 1–26%; guarantee 10% + β with β = 10%). Cost: median wall time of one 30-year forecast.

Assembly	Layout chosen	Safe rate (m²/d)	Risk on the complete assembly	Cost of one forecast
Complete assembly (referee, 25 evaluation draws)	S3.5-R3.0	6.36	≤ 10% on its own 25 draws; 11.5% expected on a new draw (Remark G2)	3.9 s
goal-guided (MONTIMI-GO), 25 design draws, no margin	S3.5-R3.0	5.80	4%	1.6 s (2.5× faster)
… with calibrated margin m = −0.15 m²/d	S3.5-R3.0	5.95	8%	
uniform coarse mesh, same terms, 25 design draws, no margin	S3.5-R3.0	5.90	4%	1.7 s (2.3× faster)
… with calibrated margin m = −0.08 m²/d	S3.5-R3.0	5.98	8%	
goal mesh, density lagged one step, 10 design draws, no margin	S2.5-R3.0	5.16	20%	0.2 s (17.8× faster)
… with calibrated margin m = −0.001 m²/d	S2.5-R3.0	5.16	20%	
goal mesh, immobile zones removed, 10 design draws, no margin	S2.5-R3.0	4.39	4%	1.8 s (2.3× faster)
… with calibrated margin m = −0.38 m²/d	S2.5-R3.0	4.77	8%	

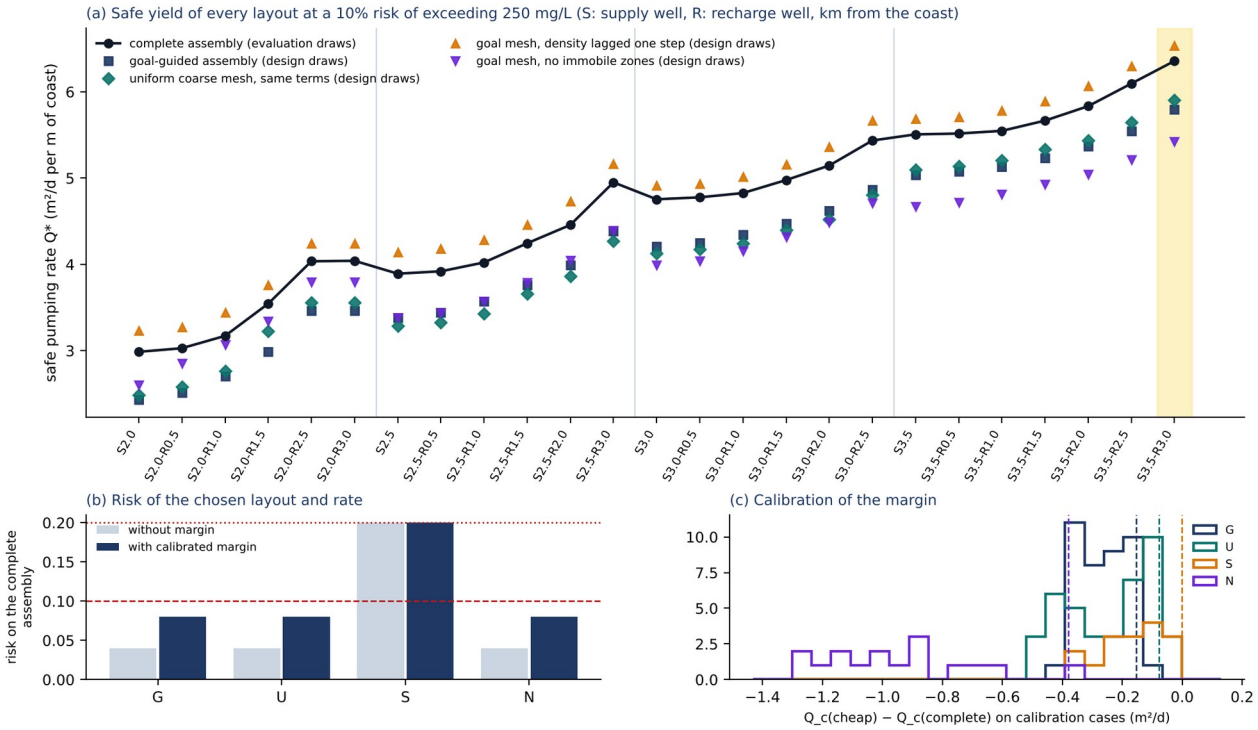


Figure 4. (a) Safe pumping rate $Q^{\star}$ of every layout on the complete assembly (black, evaluation draws) and on the cheap assemblies (design draws, no margin); the layout chosen by the complete assembly is shaded. (b) Risk of the layout and rate each cheap assembly chooses, judged on the complete assembly, without and with the calibrated margin; dashed: $\alpha = 10\,\%$, dotted: $\alpha + \beta$. (c) Calibration residuals $Q_c^{cheap} - Q_c^{complete}$ and the margins (dashed).

Both coarse assemblies are conservative: they overestimate the screen sea fraction (Table 6), so their safe rates lie below those of the complete assembly at 25 and 25 of the 25 layouts, by 0.51 m^2/d on average for the goal-guided assembly and 0.51 m^2/d for the uniform one. Yet they rank the 25 layouts almost exactly as the complete assembly does (rank correlation 0.997 and 0.999). Without a margin the goal-guided assembly chooses safely but gives water away. It chooses the supply well at 3.5 km and recharge well at 3.0 km at 5.80 m^2/d, with a risk of 4% on the complete assembly; the complete assembly would allow 6.36 m^2/d at that layout. The calibrated margins are negative: −0.15 m^2/d for the goal-guided and −0.08 m^2/d for the uniform assembly. A negative margin returns the water that the conservative bias withholds, and the guarantee of Proposition M4 still holds. With its margin the goal-guided assembly chooses the supply well at 3.5 km and recharge well at 3.0 km at 5.95 m^2/d, with a risk of 8% on the complete assembly. The uniform assembly chooses the supply well at 3.5 km and recharge well at 3.0 km at 5.98 m^2/d, with a risk of 8%. These risks rest on 25 evaluation draws: 1 of 25 (4%) has a 95% interval of 0.1–20% and 2 of 25 (8%) of 1–26%. They show that no assembly fails badly, not that the 10% target is met; the guarantee of Proposition M4 is 10% + β = 20%, and a few hundred evaluation draws of the chosen layouts would be needed to show the target itself. The complete assembly itself chooses the supply well at 3.5 km and recharge well at 3.0 km at 6.36 m^2/d. The goal-guided assembly chooses the complete assembly's own best layout. Over conveyance prices from 0 to 0.4 per km, the goal-guided assembly with its margin chooses the complete assembly's layout for 4 of 5 prices and loses at most 0.1% of the value elsewhere. The two counter-examples remove a term the goal kept, on the same 10 design draws, so they can be compared with each other. Lagging the density one step (S) removes the conservative bias of the goal mesh: the 90% calibration residual is −0.001 m^2/d against −0.15 for the goal-guided assembly, so the lag raises the critical rates by about 0.15 m^2/d, in the direction of the one-pass split error of Section 3.4. With nothing left to cover the sampling error of 10 draws, its choice, the supply well at 2.5 km and recharge well at 3.0 km at 5.16 m^2/d, exceeds the standard in 20% of the evaluation draws of the complete assembly, 2.0 times the risk allowed, with or without its margin. Without the immobile zones (N) the assembly is conservative by at least 0.38 m^2/d in nine calibration cases of ten; that covers the same sampling error (risk 4% without the margin, 8% with it), but it chooses the supply well at 2.5 km and recharge well at 3.0 km and gives up 0.7% of the value. Part of both risks comes from having only 10 design draws; the difference between the two, on the same draws, is the effect of the term removed. One 30-year forecast took 3.9 s on the complete assembly and 1.6

s on the goal-guided one, 2.5 times less; the whole placement (all layouts, all draws) took 6.1 CPU-hours on the complete assembly and 1.9 on the goal-guided one.

6.5 The toe

The toe moves as (23) says, and the relaxation law (24) describes it. Fitted to each of the 75 forecasts of the complete assembly (three layouts at their safe rates, 25 evaluation draws), with the head at mid-depth 0.5 km from the coast as the driver, it reproduces the toe over 30 years with a median largest error of 1.2 and 0.8 m for 2 of the 3 layouts, and of 20 m for the layout where the toe hardly moves (median displacement 4 m), so that there is little to fit. The median relaxation time is 7–12 years and the median γ is 556–1191 m per m of head (KUFIRI-GO assumed 8 years and 1,000 m). As a proxy for the goal the toe does better here than the sharp-interface picture suggests: a threshold on the toe position, tuned to flag as many aquifers as actually exceed 250 mg/L at the screen, catches 6 of the 7 exceedances of the three layouts. With 7 events this is thin evidence, and the toe explains only part of the variation of the screen sea fraction (correlation 0.51–0.67). The inequality (25) says why the toe cannot replace the screen: the screen meets sea water through the mixing zone and the upconing below the well, long before a sharp-interface toe would reach nine times the distance of the well, so a toe criterion must be calibrated against the screen, layout by layout.

6.6 The ions, computed once

For the first reference layout at its critical rate, the complete assembly was run again with the seven-ion block and Gaines–Thomas exchange (E = 60 meq/L) through the whole history and forecast. The goal was identical to the last digit, as Theorem M1 says. The chloride computed by the ion block agrees with the chloride of the salt block within 3.0×10^{-3} mg/L. The ion block cost 1 s of the 13 s of that run (11%). The check it is kept for is sodium. At the screen it is 125 mg/L at the end of the horizon, when chloride is 250 mg/L: below 200 mg/L, as Proposition P3 of [6] predicts for intrusion with sodium taken up by the exchanger. Hardness at the screen is 6.30 meq/L against 3.65 for the recharge water. Behind the recharge well, 100 m landward at mid-depth, the water carries 18 mg/L of sodium and 13 mg/L of chloride.

6.7 One adjoint ranks the recharge sites

The adjoint sweep of each reference case also gives the sensitivity of the screen chloride to a small recharge well at every column (53). Ranking the candidate recharge sites by this marginal value, from one forward and one adjoint run, and comparing with the ranking by the safe yield of the complete assembly with a 1 m^2/d recharge well at each site (25 draws and some 685 forecasts per supply site), the rank correlation is 0.90, 1.00, 1.00, 1.00 for the supply wells at 2.0, 2.5, 3.0, 3.5 km, and the best site is the same in 3 of 4 cases. The adjoint puts the largest marginal value of recharge landward of the supply well in 2 of 4 cases.

7. For Fushë-Kuqe: the recipe, and the limits

- **Assemble flow and salt as one block**, solved by Newton with the exact Jacobian; at Fushë-Kuqe the density coupling and the immobile zones of the silt and clay are not optional for a chloride goal (weights near the limit itself), while the hydraulic memory of the lenses can wait until a goal on heads or on the clay piezometers needs it.
- **State the goal before the model**: the chloride in the screens of the supply wells of the Gorre–Fushë-Kuqe field at the planning horizon, with the risk α the utility accepts. One forward and one adjoint run of the complete three-dimensional system at a few reference layouts then give the weights of every term and the sensitivity to recharge at every candidate site.
- **Cut the chemistry for the decision, not for the check**: the seven ions, the exchanger and the learned chemistry closure do not change the chloride goal (Theorem M1) and are run once for the chosen layout, to check sodium (Proposition P3 of the complete-system note) and the softened water behind recharge wells.
- **Calibrate the cheap model against the complete one** with a few dozen paired runs and put the margin into the chance constraint; a coarse model that is consistently conservative gets its water back, one whose error changes sign does not.
- **Rank the candidate recharge sites by one adjoint run** before running any placement; on the test it agreed with the complete placement in 3 of 4 cases.

- **Watch the placed wells with an alarm built for the new pumping**: a head-based chloride alarm learned before the decision is valid after it only if the decision is corrected through the recorded pumping and an unrecorded change of climate is certified [48].
- **Measure what the weights point to**: the inland water balance, which carries the largest weight of a missing source, and chloride profiles through the transition zone near the supply screens, which set the mixing width that decides the standard.

The limits of the test are those of a cross-section. The section is two-dimensional, so wells are lines along the coast and rates are per metre of coast; the Peaceman correction, rivers, the lagoon and a leaky base are in the assembly but not in the test. The parameters follow the Fushë-Kuqe values of the series, but the aquifer is synthetic and its present state comes from an assumed pumping history, not from assimilated data. The chance constraint was judged on 25 evaluation draws, so the realised risks are resolved to about 4%. The safe yield is an order statistic, which controls the risk on the draws used; a certificate at confidence $1-\delta$ in the Learn-then-Test sense would allow no exceedance among 25 draws at $\delta = 10\,\%$, lowering the value of the chosen layout by 11.6%, and none could be issued at $\delta = 5\,\%$ [48]. The conformal margin assumes that the decision cases are exchangeable with the calibration cases, which the selection of the best layout breaks mildly. The first-order weights are reliable for the terms that are cut and for small changes of the terms that are kept, but not for the full one-pass split, whose effect is not monotone in the lag. The dual-weighted estimate was not reliable on the uniform mesh, and the upwind goal is only piecewise differentiable where flow reverses. The prices of conveyance and recharge are illustrative. The sea-level rise was cut because the inland boundary of the section fixes the fresh inflow, so the whole head field rises with the sea and almost no water moves [49]. Where drains or a low land surface cap the water table, as on the plain of Durrës, where subsidence adds to the rise, or where pumping holds the heads below sea level, the rise steepens the gradients towards the wells and can move the wedge inland [49, 50]. The cut is therefore a property of this goal, this boundary and this horizon, not of the coast: in the three-dimensional model the relative sea level measured at each wellfield (InSAR and GNSS) stays in the assembly until its weight has been recomputed there. In the same way the sea fraction is read from chloride, which is right only where all the chloride is seawater; where it may also come from a drought that concentrates the recharge or from irrigation return flow, the ions that Theorem M1 cuts (Br/Cl, Na/Cl, nitrate, boron) carry the evidence about the source, and the placement must hold for every source that tracers cannot exclude [51]. The next step is the same procedure on the three-dimensional Fushë-Kuqe sector, with the present state from the assimilation of the chloride series of 1984–2001.

8. The system at a glance

Table 8. The assembled system at a glance, and what the placement goal needs of each block.

Block	Equations	Assembled in the test	For the chloride goal
Flow with storage and wells	(1), (4), (26)	yes (monolithic with salt)	kept
Memory of the lenses (kernel, tail, lift)	(2), (3)	yes	cut (weight below 2×10^{-5})
Learned closure Ψ	(6)	zero	sensitivity mapped (Section 6.2)
Density and buoyancy	(5)	yes	kept
Salt with immobile zones and dispersion	(7)–(10)	yes	kept
Screen chloride and the standard	(11), (12)	yes (the goal)	goal
Seven ions, exchange, reactions, Φ	(13)–(21)	yes, once (Section 6.6)	cut exactly (Theorem M1); checked after
Electro-diffusion	(18)	no	cut (leak bound 4×10^{-6})
Toe and its mobility	(22)–(25)	diagnostic	not a goal (Section 6.5)
Sea boundary, sea-level rise, tides	(29), (30)	hydrostatic sea, rise	rise cut for this horizon (weight below 10^{-5})
Land, recharge, rivers, base	(31), (32)	regional inflow; closed top and base	kept (inflow); others not in the test
Initial state	(33)–(35)	pre-development + 50-year history	kept
Placement problem and adjoint	(47)–(53)	yes	the final goal

Prepared on 27 October 2026 from the code and results of experiments/montimi_go in the author's repository (stages 0–5 and the analysis); every number about the test is read from experiments/montimi_go/res and checked when the document is built. The references of other authors were checked against Crossref, the Official Journal or

library catalogues (the eleven formerly marked "not verified" on 8 November 2026); the notes of the author's series ([2]–[12], [22], [23], [48], [51]) are unpublished and available from the author.

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